Evaluation of the Diversification, Hedging, and Safe Haven Potential of Commodities for Stock Portfolios based on Dynamic, Conditional Correlations

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List of Abbreviations

DCC ..... Dynamic Conditional Correlation
DJCI ..... Dow Jones Commodity index
GARCH ..... Generalized Autoregressive Conditional Heteroskedasticity
GFC ..... Global Financial Crisis
GSCI ..... S&P GSCI index
MSCI ..... MSCI World index
MSCIEM ..... MSCI Emerging Markets index
SPGSAG ..... S&P GSCI Agriculture index
SPGSEN ..... S&P GSCI Energy index
SPGSGC ..... S&P GSCI Gold index
SPGSIN ..... S&P GSCI Industrial Metals index
SPGSLE ..... S&P GSCI Light Energy index
SPGSLV ..... S&P GSCI Livestock index
SPGSNE ..... S&P GSCI Non Energy index
SPGSPM ..... S&P GSCI Precious Metals index

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1 Introduction

Commodities are commonly believed to be attractive additions to portfolios for three reasons: low correlation with oher asset classes (primarily equities), relatively high returns (in the same range as equity returns), and positive correlation with inflation (Ali et al., 2020; Nguyen et al., 2020). Beginning in the 2000s, however, commodity markets have become subject to an increasing influx of financial investors – a process named the financialization of commodity markets (Basak & Pavlova, 2016). Some scholars advance that this has caused significant price impact, changing the market dynamics and challenging commodities' potential benefits (Basak & Pavlova, 2016; Büyükşahin & Robe, 2014; Singleton, 2014; Tang & Xiong, 2012). Nonetheless, this has been a matter of debate. Arguments speaking against a change in price behavior can be found in the work of Hamilton & Wu (2014), Irwin & Sanders (2011), and Kilian & Murphy (2014).

The influx of financial investors was accelerated by various forces. First, the inflow of investment capital was boosted by the passing of the Commodity Futures Modernization Act in 2000, which deregulated the market in many ways (Ali et al., 2020). This allowed hedge funds as well as financial institutions to trade commodity derivatives with more flexibility and prompted an extended offer of new investment vehicles (Büyüksahin et al., 2009). The second most important driver, according to Chunhachinda et al. (2019), was that the Global Financial Crisis (GFC) in 2008 motivated institutional as well as retail investors to increase their commodity investments due to the belief that commodities are uncorrelated with stock and bond prices.

To put the question into a quantitative perspective, a report by the Commodity Futures Trading Commission (2008) estimates that the aggregate net amount of all commodity index trading (combined OTC and on-exchange activity) between 2000 and June 2008 was equal to $200 billion. For comparison, the sum of investments into vehicles tracking the most prominent commodity index, the S&P GSCI index, was estimated at less than $5 billion between 1991 and 1999 (Büyüksahin et al., 2009). The development after 2008 is reflected in the rise of trading volume. According to the Futures Industry Association (n.d.), the volume of globally traded commodity futures contracts amounted to 1.9 billion in 2008. In 2023, the number of traded contracts was 11 billion (Futures Industry Association, n.d.).

The main concern about the financialization is that it results in commodity prices resembling the behavior of other financial asset classes. This would suggest a stronger positive correlation with stock prices (Nguyen et al., 2020). In particular, this could be caused by differences in the behavior of commercial traders (traditional commodity traders), who use commodity futures to hedge the spot-price risk of their commercial activities, and the behavior of financial institutions. For example, financial investors could be forced to liquidate their position in a commodity due to a decline in the stock market resulting in a need for cash for margin calls (Büyüksahin et al., 2009). Thereby, stock market volatility could spill over to commodity markets. Ali et al. (2020) suggest that the consequences would include an elevation of prices, volatility, and correlation among commodity classes. In conclusion, financialization could inhibit the hedging properties of commodities.

Ultimately, this raises the question whether commodities are effective hedges, safe havens, or diversifiers. This thesis addresses the hedging, safe haven and diversification potential of commodities for stock portfolios between 1992 and 2023.

The practice of diversification is defined as allocating the investment amount across different assets in order to reduce risk by minimizing the probability as well as severity of losses (Koumou, 2020; Lhabitant, 2017). The beginning of academic research on diversification is marked by the work of Markowitz (1952). According to Markowitz (1952), one important feature of a diversifier, in comparison to a hedge or safe haven, is that it is not supposed or able to eliminate all variance. Nonetheless, it should be underlined that a diversified portfolio has to consist of securities with low covariances among themselves in order to reduce the risk of simultaneous poor performance of all securities. Eventually, the purpose of a diversifier is to achieve a higher risk-return ratio than the undiversified portfolio, even if this leads to a decrease of expected return (Markowitz, 1952).

In comparison to diversifiers, assets considered as hedges or safe havens are expected to have even lower co-movement, exposure to traditional risk factors, sensitivity to market returns, and volatility (Kopyl & Lee, 2016). Hedges and safe havens differ in the way that hedges do, on average, not move in line with the hedged asset, but may co-move during periods of severe market stress due to herding or contagion effects (Kopyl & Lee, 2016). Conversely, safe havens are characterized by their low correlation during times of crises.

Their average behavior during normal, non-crisis market periods is irrelevant (Kopyl & Lee, 2016). Concluding, hedges may be viewed as "long-term investments to reduce portfolio risk and fluctuations in returns" (Kopyl & Lee, 2016, p. 457), while safe haven investments are supposed to increase the portfolio performance specifically in periods of market turmoil.

Following the approaches of Ali et al. (2020) and Nguyen et al. (2020), the three terms will be defined based on Baur and Lucey (2010) in this thesis. This framework is widely used in research on portfolio diversification. It has been applied to a number of assets, including gold (Baur & Lucey, 2010), credit default swaps (Ratner & Chiu, 2013), and bitcoin (Bouri et al., 2017). The definitions are as follows:

> "A hedge is defined as an asset that is uncorrelated or negatively correlated with another asset or portfolio on average. A hedge does not have the (specific) property of reducing losses in times of market stress or turmoil since the asset could exhibit a positive correlation in such periods and a negative correlation in normal times with a negative correlation on average. […] A diversifier is defined as an asset that is positively (but not perfectly correlated) with another asset or portfolio on average. Similar to the hedge, the diversifier does not have the (specific) property of reducing losses in extreme adverse market conditions since the correlation property is only required to hold on average. […] A safe haven is defined as an asset that is uncorrelated or negatively correlated with another asset or portfolio in times of market stress or turmoil. The specific property of a safe haven asset is the nonpositive correlation with a portfolio in extreme market conditions. This property does not force the correlation to be positive or negative on average but only to be zero or negative in specific periods. Hence, in normal times or bullish market conditions the correlation can be positive or negative." (Baur & Lucey, 2010, p. 219)

The remainder of this paper is structured as follows: Section 2 gives an overview of the existing literature on the topic. Section 3 is dedicated to the employed methodology and contains subsections focused on the data, descriptive statistics, and the DCC GARCH framework. Subsequently, the results are discussed in Section 4. First, the model parameters will be explored in 4.1. Then, in Sections 4.2 and 4.3, the correlation of the GSCI and S&P 500 will be analyzed with emphasis on three events: the GFC, the COVID

Market Crash, and the Russian invasion of Ukraine. In the successive subsection, the implications for the classification as diversifier, hedge, or safe haven are elaborated. In Section 4.5, the individual commodity subclasses are assessed, and in Section 4.6, the findings of robustness checks presented. Finally, Section 5 concludes.

The results show that composite commodity indices are neither effective hedges nor safe havens. They could be considered as diversifiers, though the average correlation with stocks is relatively high – for the S&P 500 and GSCI, the average correlation is 0.329. Regarding safe haven properties, it should be mentioned that composite commodity indices acted as safe havens during minor but not major market crises. Furthermore, this thesis finds that the GFC in connection with the financialization had a major impact on the correlation levels, shifting them into a positive direction. Conversely, COVID and the Russian invasion led to different reactions. While COVID saw higher correlations, the correlation decreased in the aftermath of the invasion. Turning to commodity subindices - energy, industrial metals, and livestock could serve as diversifiers. Gold and precious metals are found to be appropriate hedges, as well as safe havens.

The findings are relevant for investors, portfolio managers, and academics. For investors and portfolio managers, this research provides valuable information about the potential of commodity investments. Therefore, it supports them in making informed decisions about incorporating commodities into their portfolio strategies. Academics can use the conclusions as a starting point for further research. This applies particularly to the understudied recent crises between 2020 and 2023.

2 Literature Review

Due to the increasing popularity of commodities as investment vehicles, there has been a growing amount of academic work on the value commodities add to stock portfolios. This section gives an overview of the most important works in the field. Generally, the majority of research on the topic either focuses on the correlation between commodities and stocks (see Büyüksahin et al., 2009; Creti et al., 2013; Garcia-Feijoo et al., 2012; Junttila et al., 2018; Lombardi & Ravazzolo, 2016; Nguyen et al., 2015, 2020; Silvennoinen & Thorp, 2013) or tracks the performance of commodity-hedged portfolios in terms of risk and return (see Bessler & Wolff, 2015; Chunhachinda et al., 2019; Conover et al., 2010; Daskalaki & Skiadopoulos, 2011; Olson et al., 2017). Moreover, the large body of research on the individual commodities of gold and oil should be highlighted (see Baur & Lucey, 2010; Baur & McDermott, 2010; Choi & Hammoudeh, 2010; El Hedi Arouri et al., 2011, 2015; Filis et al., 2011; Junttila et al., 2018).

Considering the scope of this thesis as well as the overweight of research on correlation in contrast to portfolio performance, the following literature review will concentrate on papers centered around correlation properties. Furthermore, it will be focused on studies analyzing composite commodity indices as well as different subclasses, instead of solely individual commodities such as gold. Thirdly, only papers written after the mid-2000s will be considered. Given the changes in availability and use of commodities as investment vehicles, research conducted before this time hold only low informational value for the present (for earlier studies see Ankrim & Hensel, 1993; Bodie, 1983; Jensen et al., 2000).

Büyüksahin et al. (2009) were among the first scholars publishing in the field after the financialization. Therefore, their work is fundamental and referenced in most of the literature. Büyüksahin et al. (2009) implement a DCC GARCH model to estimate the correlations for short time horizons. DCC GARCH stands for Dynamic Conditional Correlation Generalized Autoregressive Conditional Heteroskedasticity. It is a multivariate volatility model that captures time-varying correlations among different time series by allowing their conditional correlation to evolve over time and will also be computed in this thesis. For long-term co-movements, Büyüksahin et al. (2009) perform a (Recursive) Cointegration Analysis. In contrast to this, later published papers rely more frequently only on GARCH model correlations without conducting further, alternative analyses (see Creti

et al., 2013; Junttila et al., 2018; Lombardi & Ravazzolo, 2016; Silvennoinen & Thorp, 2013).

Büyüksahin et al. (2009) investigate the time period from June 1992 to June 2008 with a focus on establishing if the relation between the returns of commodity indices and equity indices has changed during that time as a consequence of financialization. Their results show that there is no structural shift visible between 1992 and 2008 - neither during calm market times, nor during market distress. The correlation itself is described as strongly fluctuating. For comprehensive commodity indices, they find that correlations are frequently below 0.3 or even negative. Furthermore, they underline the differences between non-energy and energy-including indices as well as the low equity correlation of narrow commodity classes. This observation stresses the importance of sampling a variety of commodity (sub)indices.

For extreme events, the authors conclude that, even though the correlations strengthen, there is only little evidence of a linkage between extreme equity returns and commodity returns. Subsequently, they establish that "investors can still achieve substantial gains by diversifying portfolios across the two asset classes" (Büyüksahin et al., 2009, p. 4). When classifying commodities based on the correlations found by Büyüksahin et al. (2009), they may be viewed as diversifiers or hedges. It has to be considered that the GFC with its full length is not included in the sample. This is crucial since many authors propose that the GFC led to a reinforcement of financialization, thereby changing the relations between commodities and equities (Basak & Pavlova, 2016; Creti et al., 2013; Lombardi & Ravazzolo, 2016; Nguyen et al., 2020; Tang & Xiong, 2012). Therefore, the results may be regarded as an accurate evaluation of the past but may not be applicable to the present.

The research of Garcia-Feijoo et al. (2012) corroborates the necessity of differentiating between commodity classes. In their study, the hedging effectiveness of a variety of asset classes, including a commodity index and subindices, was investigated for the time period from 1970 to 2011. For all commodities, the authors establish an ability to hedge against extreme market shocks, especially in inflationary environments. This is indicated by the commodity index performance not significantly deviating from its average level in the times of extreme adversity in stock markets (Garcia-Feijoo et al., 2012). In spite of that, it has to be taken into account that the performance of the subclasses differs. For example, industrial metals are more strongly correlated with equities, thereby offering less hedging

effectiveness, and precious metals as well as energy prove most effective in terms of inflation. The same is found by Erb & Harvey (2006), who argue that commodities cannot be treated as a single asset class given that they establish an average correlation of only 0.09 across individual commodity classes.

Regarding composite commodity indices, Garcia-Feijoo et al. (2012) find an insignificant stock-correlation for earlier periods, though the correlation of 0.13 for the later period of 1990 to 2010 is significant. Similar to Tang and Xiong (2012), the authors attribute this change to the financialization, thereby contradicting the results of Büyüksahin et al. (2009). It should, however, be pointed out that the work of Garcia-Feijoo et al. (2012) is based on the Pearson Correlation Coefficient, which does not capture the time-varying nature of correlations and is most probably less accurate than the DCC GARCH correlation implemented by Büyüksahin et al. (2009). Secondly, the difference in findings could be due to the longer time period investigated by Garcia-Feijoo et al. (2012). Their research may capture the impact of the GFC to greater extent. Concluding, composite commodity indices would be classified as hedges for earlier periods and as diversifiers for periods after the financialization, based on Garcia-Feijoo et al. (2012).

Creti et al. (2013) add to the existing literature about the heterogeneity of commodities by not investigating indices, but a large number of individual commodities. More precisely, their data includes daily spot prices of 25 raw materials in the sectors of energy, precious metals, agricultural, non-ferrous metals, food, oleaginous, exotic, and livestock from 2001 to 2011. The commodities' relevance is ensured by "important trading activity, as apprehended through the transaction volume and the number of large participants in the market" (Creti et al., 2013, p. 19). Similar to Garcia-Feijoo et al. (2012), they find a structural shift in correlations during the GFC in 2008 and explain this with financialization.

For all commodities, Creti et al. (2013) observe highly volatile correlations, especially during and after the GFC. Secondly, the largest decrease appears during the GFC for all commodities. According to the paper, the crisis did, however, only loosen the conditional links in the short run. In the long run, it caused a significant increase of correlations. Creti et al. (2013) conclude that the correlations are driven by industrialization, financialization, and commercial as well as non-commercial traders.

In contrast to Garcia-Feijoo et al. (2012), the correlations of Creti et al. (2013) are computed with a DCC GARCH model instead of Pearson. Consequently, the time-varying nature of correlations is captured. In this way, the findings of Garcia-Feijoo et al. (2012) are corroborated by the extension to a variety of individual commodities as well as the use of a sophisticated DCC GARCH model. Furthermore, Creti et al. (2013) support the conclusion of Garcia-Feijoo et al. (2012) and Büyüksahin et al. (2009) that commodities cannot be viewed as a homogenous asset class. They find a speculation phenomenon for oil, coffee, and cocoa, and safe haven properties for gold.

Oil is presented as the commodity class most closely related to the stock market (for similar results see Choi & Hammoudeh, 2010 and Filis et al., 2011). Creti et al. (2013) explain this with the fact that an increase in oil prices results in higher production costs for many companies, ultimately limiting profits and causing a decrease in shareholders' value, which becomes visible in stock prices. In periods of increasing stock prices, the correlations between stocks and oil tend to rise. Conversely, during times of declining stock prices, these correlations typically decrease. Correlations included an equal amount of both positive and negative values before 2008, but remained positive between 2008 and 2012.

This pattern aligns with the well-documented phenomenon of oil speculation, which states that the surge in crude prices is heightened during bull markets (Creti et al., 2013). The oil-equity correlations remain positive for the majority of time after the GFC. From this standpoint, oil is not considered a viable means of diversification, hedging, or safe haven. As of other energy commodities, gas and electricity exhibit rising correlations at the beginning of the observed period and then stabilize, unaffected by stock market conditions.

Notably, negative correlations are often observed between stock and electricity markets, suggesting that especially electricity could serve as a hedge. Similar to oil, this may be based on the grounds that an increase of electricity prices results in higher production costs for companies, which in turn negatively impacts their stock price. Creti et al. (2013), on the other hand, advance that the negative correlation indicates that the “electricity market is mainly driven by its own market fundamentals (i.e. non-storability, inelasticity of the supply…)” (Creti et al., 2013, p. 20).

For gold as well as precious metals in general, Creti et al. (2013) find mostly negative correlations, which decline further during periods of diminishing stock prices. Therefore, the authors classify them as safe havens, supporting Baur & McDermott (2010) and Choi & Hammoudeh (2010). Regarding the group of exotic commodities, the correlation of sugar does not react to different stock market conditions, and hovers around a positive value of 0.1, constituting it a diversifier. Coffee and cocoa do, however, show the same pattern as oil with growing correlations during times of increasing stock prices and diminishing correlations during times of declining stock prices. Additionally, they have shifted to relatively high correlation levels of around 0.3 after the GFC. Consequently, other commodities, such as sugar, may be regarded as better diversifiers.

In contrast to the previously mentioned papers, Nguyen et al. (2020) do not compute the general correlation, but the correlation of unexpected variations (e.g. caused by shocks) in equity returns and unexpected variations in commodity future returns. As a foundation for distinguishing between expected and unexpected variations, the best data-generating process for equity and commodity future returns is estimated. The reasoning for this approach is that the correlation between simple returns does not account for the difference of systematic and idiosyncratic components.

The systematic component represents the part of the correlation that is driven by broad market trends and factors affecting the entire market, whereas the idiosyncratic component represents the part of the correlation that is specific to individual assets (Nguyen et al., 2020). While the simple return correlation includes the systematic component, the correlation between shocks to returns is supposed to only capture the idiosyncratic component (Nguyen et al., 2020). Given the focus on shocks, their work is especially valuable for evaluating the safe haven potential of commodities.

For their sample from 1990 to 2017, they find a low, significant, positive correlation between equity return shocks and composite commodity index return shocks. Subsequently, according to Nguyen et al. (2020), diversified commodities do not serve as a safe haven. Instead, this implies that commodities may be diversifiers. Nonetheless, a negative, though insignificant, correlation between equity return shocks and gold return shocks is established. Hence, gold may be viewed as a safe haven, corroborating the results of Creti et al. (2013).

For other subclasses, the correlations range from 0.131 to 0.331, wherefore they are not suitable as safe havens, but may be diversifiers. Robustness checks show that the safe haven properties differ for distinct time horizons. For example, gold can only be evaluated as a safe haven for short-term variations, but not mid-term fluctuations, meaning it should be bought and hold for time horizons below one year (Nguyen et al., 2020).

Additionally, the results suggest that the financialization has had an impact on correlations. Nguyen et al. (2020) split the sample into two sub-samples, one from 1990 to 1999 and one from 2000 to 2017. In contrast to the results for the complete time period, all correlations for the first sub-sample are lower (some even negative), and insignificant. The only exception is crude oil, which has a negative correlation at a significance level of 10%. For the second sub-sample, crude oil does, however, exhibit a positive correlation.

This holds true for the other commodities, which all have higher, positive, and mostly significant correlations in the second sub-sample. Nguyen et al. (2020) conclude that the shift of correlation levels may be due to financialization. In summary, oil could have been used as a safe haven during the 90s - for the other commodity classes, further research would have to be conducted to evaluate their potential during the 90s. For 2000 to 2017, it can be concluded that they only have potential as diversifiers.

Lombardi & Ravazzolo (2016) extend the literature by using the DCC GARCH model in an asset-allocation framework. Their findings suggest that employing a time-varying joint model for commodity and equity prices yields economic benefits compared to a passive strategy of holding 100% in equities, especially during periods of significant price fluctuations. This means that the simulated stock-commodity-portfolio produces a higher Sharpe Ratio than the stock portfolio. The largest gains arise during the GFC. Simultaneously, an investment strategy that incorporates commodities in a portfolio results in substantially higher volatility compared to portfolios that exclude them. This heightened volatility is evident for short investment horizons throughout the entire sample period from 1980 until 2015.

When evaluating the portfolio value of commodities, it is, therefore, crucial to define if a hedge or diversifier is supposed to reduce the risk as measured by volatility or if it is supposed to result in a higher risk-return ratio as measured by the Sharpe Ratio. The definition of Baur & Lucey (2010), which serves as the basis for this thesis, does not

include such considerations. Lombardi & Ravazzolo (2016) conclude that the increase in volatility contradicts the assumption that commodities could serve as a hedge. It should be noted that the study of Lombardi & Ravazzolo (2016) is based solely only the GSCI. Their results may not be applicable to other indices or subclasses.

On the subject of correlations, Lombardi & Ravazzolo (2016) observe both positive and negative trends. On average, they find correlations remaining close to zero. In the recent period, however, correlations have shifted to higher, positive levels compared to historical patterns. This shift commenced with the onset of the GFC in 2008, and, although correlations have decreased since then, they still maintain a positive direction, stabilizing at a level of around 0.25, meaning somewhat higher than the pre-recession period. Based on the methodology of Baur & Lucey (2010), the results of Lombardi & Ravazzolo (2016) suggest that the GSCI could have served as a hedge for the time prior to the GFC. Since then, it could only be added to a portfolio as a diversifier.

This thesis adds value to the existing body of academic work by including recent data into the analysis. Most importantly, the sample includes the recent market crises due to COVID and the Russian war against Ukraine, as well as the highly inflationary period of 2022. Given the long period of economic growth before 2020, this yields important insights into if commodity performance has changed in times of market distress. Up until this date, few papers have been published on the subject after 2020. All of the mentioned research relies mostly on the GFC in 2008 to evaluate the properties of commodities during market turmoil. Considering the special role of the GFC in the context of financialization, it is important to investigate the correlation between stocks and commodities in other crises to ensure generalizability.

3 Methodology

This section is dedicated to elaborating on the methodology of the data analysis. First, the used data and indices are presented. Then, the descriptive statistics are computed to give an overview of the indices and their performance properties. Lastly, a DCC GARCH framework is used to establish the time-varying correlation between stocks and commodities. The correlation is analyzed separately for two subsamples to describe the relationship during special market times as well as to ensure the generalizability of the results. As a robustness check, different stock and commodity indices as well as commodity subclasses will be tested.

3.1 Data

The empirical analysis is based upon daily data extracted from Bloomberg. The S&P 500 will be the primary equity index. The S&P 500 is a stock market index that measures the performance of the 500 largest publicly traded companies in the United States and covers approximately 80% of available US market capitalization (S&P Global, n.d.-b). It is one of the most widely followed equity indices in the world and is considered a reliable indicator of the overall health and performance of the stock market by a great number of funds, regulators, and companies (Robertson, 2023).

Given that the S&P 500 only contains large-cap US stocks, it has, however, been pointed out that the index may not always be an adequate benchmark (Robertson, 2023). This is why the MSCI World index (MSCI) will be used to verify the reliability of the results for stock markets overall. Apart from the S&P 500, the MSCI is the most recognized global equity index. It measures the performance of large- and mid-cap stocks from 23 developed countries around the world, capturing approximately 85% of market capitalization in each country (MSCI, n.d.-a).

Both indices are capitalization-weighted, meaning that large-cap companies have a more significant influence on the index performance (MSCI, n.d.-a; S&P Global, n.d.-b). The assessment of indices consisting primarily of small- and mid-cap stocks is beyond the scope of this thesis. Furthermore, both indices are limited to developed markets. Chunhachinda et al. (2019) demonstrate that the inclusion of commodities in emerging market stock portfolios leads to greater risk reduction than in developed market portfolios. Accordingly, the analysis of this thesis will contain the MSCI Emerging Markets index

(MSCIEM) to account for emerging markets. The MSCIEM consists of large- and mid-cap stocks across 24 emerging markets – it is also capitalization-weighted (MSCI, n.d.-b).

For commodities, the S&P GSCI index (GSCI) will be used. The GSCI is the most widely used benchmark for commodity investments (see for example Bessler & Wolff, 2015; Büyüksahin et al., 2009; Chunhachinda et al., 2019; Garcia-Feijoo et al., 2012; Lombardi & Ravazzolo, 2016; Nguyen et al., 2015, 2020). It consists of production-weighted, long-only investments in futures across a variety of commodity classes (S&P Global, n.d.-c). The included subclasses are Energy, Livestock, Industrial Metals, Precious Metals, Gold, and Agriculture. Even though the GSCI contains different commodities, it has a strong overweight of energy futures.

Therefore, the S&P GSCI Light Energy (SPGSLE), as well as the Dow Jones Commodity index (DJCI), will be analyzed as possible alternatives for the sake of robustness. The SPGSLE contains the same designated contracts as the GSCI, but with a modification in the contract production weights for the energy sector. The weights for energy contracts are divided by four, resulting in a reduction of their relative influence. This, in turn, amplifies the significance of non-energy constituents within the SPGSLE, ultimately shaping a more diversified benchmark that reflects the broad performance of the commodity market (S&P Global, n.d.-a). The DJCI follows liquidity weighting (measured by Total Dollar Value Traded) with a cap of 33% to ensure that all three sectors (Energy, Agriculture and Livestock, All Metals) are equal-weighted (S&P Global, n.d.-a). Apart from the GSCI, the DJCI is the most widely followed commodity index.

Moreover, the individual commodity subclasses of the GSCI will be considered with the respective indices. This is appropriate in the face of research demonstrating low correlation between commodity classes (Ali et al., 2020; Nguyen et al., 2020; Olson et al., 2017). The subindices are the S&P GSCI Energy (SPGSEN), S&P GSCI Livestock (SPGSLV), S&P GSCI Industrial Metals (SPGSIN), S&P GSCI Precious Metals (SPGSPM), S&P GSCI Gold (SPGSGC), and S&P GSCI Agriculture (SPGSAG). Of these indices, only the SPGSAG will not be included due to the lack of data availability on Bloomberg.

It should be noted that this thesis is based on the daily returns r_t of commodities indices calculated in the following way:

$$r_t = \frac{p_t}{p_{t-1}} \tag{1}$$

where p_t is the closing price of the day.

This does, however, not depict reality with full accuracy. For investors, commodity returns consist of three components: spot return, cash or collateral return, and roll return (Ang, 2014). The spot return is equal to the price change in the spot market – this is the return analyzed in this thesis. The cash return is earned by interest payments on the required collateral for the futures contract. Lastly, the roll return is generated by rolling over to new futures contracts as the old ones reach expiration to gain continuous exposure. While spot prices are influenced by production factors and commodity investment activities, the cash return is determined by interest rate factors, and the roll return by the slope of the futures curve (Ang, 2014).

Both the cash and roll components significantly contribute to the overall return of commodities and the relative contributions of each return are subject to change. Nonetheless, the primary source of variance and risk in commodity future returns are fluctuations in underlying spot markets (Ang, 2014). Considering that cash and roll returns constitute a minor source of variance, this thesis will be restricted to the use of spot returns. More specifically, the S&P GSCI Index Spot Index of Bloomberg was used.

The sample period ranges from January 1, 1991 to September 15, 2023. It covers recessions, such as the GFC, as well as times of economic growth, such as the period from 2010 to 2020. The beginning of the sample is marked by the creation of the GSCI by Goldman Sachs in 1991 (S&P Global, n.d.-c). It should be noted that the first contract on the GSCI was offered on the Chicago Mercantile Exchange in July 1992, though the performance history was backfilled to 1969 (Erb & Harvey, 2006). Hence, Erb and Harvey (2006) suggest that the performance prior to July 1992 should be treated with caution.

In this thesis, the data from January 1991 to July 1992 will be used to compute correlations within the DCC GARCH framework. Due to the high number of data points lowering the influence of the backfilled data, as well as the time-varying nature of DCC correlations, which allows to evaluate how the correlation evolved after the backfilled period, this approach is deemed appropriate. It does not overweigh the backfilled data, but still leverages its informational value.

To evaluate the correlation in different market environments, the period will be divided into two sub-samples when describing the results. The first sample is from January 1991 until December 2010 and will be examined with emphasis on the financialization during the 2000s and the GFC. The second sample covers January 2011 until September 15, 2023. Therefore, its analysis will mainly revolve around the crises due to COVID Market Crash and the Russian invasion.

Bessler and Wolff (2015) strongly suggest out-of-sample analysis. In their comparison of out-of-sample and in-sample analyses, they found that previous in-sample studies have exaggerated the advantages of commodities for stock portfolios to great extent. Out-of-sample analyses also have a higher applicability to the real world considering that investors only possess the historical information available at the time of their portfolio allocation decision. They cannot include future information in their evaluation.

Therefore, an out-of-sample approach will be followed in this thesis. Only past information will be used to compute the correlations. The input values will be all historical data points from the beginning of the overall sample in 1991. It was decided against rolling windows since all information should be leveraged and the DCC GARCH framework allows to overweigh the recent data.

3.2 Descriptive Statistics

The following section elaborates on the descriptive statistics of the daily returns to give an overview of the data characteristics and is based on the approaches of comparable papers in the field (see Ali et al., 2020; Bessler & Wolff, 2015; Büyüksahin et al., 2009; Chunhachinda et al., 2019; Daskalaki & Skiadopoulos, 2011). More precisely, the mean, median, maximum, minimum, standard deviation, skewness, kurtosis, and Jarque-Bera p-value are computed for the daily returns of all indices during the entire sample period. Furthermore, a simple correlation matrix is estimated.

	Mean	Median	Maximum	Minimum	SD	n
S&P 500	0.00028	0.00058	0.116	-0.120	0.0124	8239
MSCI	0.00020	0.00062	0.095	-0.099	0.0103	8533
MSCIEM	0.00025	0.00083	0.106	-0.095	0.0117	8533
GSCI	0.00041	0.00085	0.080	-0.118	0.0146	8246
DJCI	0.00032	0.00049	0.067	-0.081	0.0101	6218
SPGSLE	0.00028	0.00047	0.064	-0.070	0.0102	8247
SPGSEN	0.00064	0.00109	0.190	-0.260	0.0212	8248
SPGSGC	0.00032	0.00035	0.092	-0.093	0.0109	8247
SPGSIN	0.00027	0.00016	0.079	-0.087	0.0134	8248
SPGSLV	0.00017	0.00024	0.054	-0.060	0.0096	8248
SPGSPM	0.00032	0.00041	0.092	-0.096	0.0114	8248

Table 1. Descriptive statistics of daily returns between January 1, 1991 and September 15, 2023

Table 1 shows the mean, median, maximum, minimum, and standard deviation of the returns. Moreover, the table gives the number of observations. The small differences in number of observations are due to inconsistencies in price reporting – for some indices, the prices for holidays or weekends are reported as the price of the preceding trading day. Other datasets do not include any holidays or weekends. Those discrepancies are adjusted in the analysis.

Regarding the DJCI, Bloomberg only provides prices after the 8th of January, 1999, causing a smaller number of observations. Additionally, it should be noted that Büyüksahin et al. (2009) report substantial performance differences of commodity indices for varying time periods. This leads to a strong impact of the chosen time horizon on descriptive statistics. For example, the authors point out that both industrial and precious metals exhibit lower performance compared to all other commodity sub-indices from 1992 to 1997. In spite of that, they outperform all except for energy from 2003 to 2008. Subsequently, the following results should be interpreted in the face of the time period.

As of equity indices, the mean returns range from 0.020% to 0.028% with maximum values from 9.5% to 11.5% and minimum values from -12.0% to -9.5%. The median values vary more greatly. While the S&P 500 has a median return of 0.058%, the MSCIEM

produces a median return of 0.083%. Nonetheless, it can be stated that the stock indices do not perform significantly differently, based on the descriptive statistics.

Turning to composite commodity indices, the GSCI yields an average return of 0.041% and a median return of 0.085%, whereas the DJCI demonstrates an average of 0.032% and a median of 0.049%. The performance of the SPGSLE rather resembles the DJCI with a mean of 0.028% and a median of 0.047%. This observation corroborates the assumption that the GSCI performance is influenced by its overweight of energy futures and demonstrates the benefits of using the DJCI and SPGSLE as alternative benchmarks for robustness. Looking at average returns, the commodity indices reach the same or even slightly higher levels than stock indices, depending on the measure (mean or median). On the other hand, equity returns display higher maximum and minimum values.

For commodity subindices, the average and median returns range from 0.017% to 0.064% and from 0.016% to 0.109%, respectively. The maximum returns can be found between 19.0% and 5.4% and minimum returns between -26.0% and -6.0%. This shows that commodities are a heterogenous asset class and proves the necessity of analyzing the subclasses individually.

The SPGSEN can be considered the high performer, while the SPGSLV has the lowest average and the SPGSIN the lowest median return. It should, however, be taken into account that the SPGSEN also has the highest maximum and lowest minimum, hinting at increased volatility, risk, and possibility of extreme movements. These findings are not in line with other research due to the strong dependency on the chosen time horizon. For example, Ali et al. (2020) suggest that gold outperformed energy commodities between 2001 and 2018 in terms of daily returns.

Taking a look at standard deviations, it becomes evident that especially the GSCI and SPGSEN are volatile assets with standard deviations of 0.0146 and 0.0212, respectively. This proves again the overweight of energy assets in the GSCI. Nevertheless, the two assets also have the highest returns of all assets, indicating that the risk is rewarded.

Although commodities are known for their volatility (Baur & Dimpfl, 2018; Brooks & Prokopczuk, 2013; Pindyck, 2004), the other commodity assets do not display higher standard deviations than the stock indices, though they show just as great discrepancies in standard deviations as in returns. In fact, the SPGSLV even has the lowest standard

deviation of all analyzed indices with 0.0096. In contrast to this, Daskalaki and Skiadopoulos (2011) report greater standard deviations and lower monthly returns for commodity indices in comparison to stock indices between 1989 and 2009.

	Skewness	Kurtosis	Jarque-Bera (p-value)
S&P 500	-0.136	12.992	0.0000
MSCI	-0.372	13.053	0.0000
MSCIEM	-0.315	10.522	0.0000
GSCI	-0.289	7.217	0.0000
DJCI	-0.369	7.010	0.0000
SPGSLE	-0.299	6.720	0.0000
SPGSEN	-0.291	13.358	0.0000
SPGSGC	-0.021	9.442	0.0000
SPGSIN	-0.132	5.964	0.0000
SPGSLV	-0.125	4.696	0.0000
SPGSPM	-0.171	9.119	0.0000

Table 2. Descriptive statistics on distribution features of daily returns between January 1, 1991 and September 15, 2023

Resembling the results of research on the indices for other time and return periods, Table 2 demonstrates that all time series are negatively skewed and possess high values for kurtosis (Ali et al., 2020; Daskalaki & Skiadopoulos, 2011; Nguyen et al., 2020). Negative skewness suggests that, in the past, investors have experienced frequent small gains and few large losses. High kurtosis, on the other hand, conveys that there have been many price fluctuations in the past (positive or negative) away from the average returns. The coefficients already hint at the fact that the returns are not normally distributed. This assumption is confirmed by the highly significant Jarque–Bera test, which rejects the null hypothesis of normal distribution for every index at a 1% significance level.

When comparing the distinct assets, the low absolute value of skewness of the S&P 500 compared to the other stock indices is notable. Except for that, the composite commodity indices and stock indices have similar values. Concerning the individual commodity classes, stronger differences are visible. The SPGSEN has an exceptional strong skewness of -0.291, whereas the SPGSGC has a remarkably low skewness of -0.021.

Regarding kurtosis, the stock indices display substantially higher values than commodities. The only exception is the SPGSEN with a kurtosis of 13.358, corroborating the earlier findings of high volatility of this index. The greater the skewness and kurtosis values, the more pronounced are the features related to negative skewness and high kurtosis mentioned above.

	S&P 500	MSCI	MSCIEM	GSCI	DJCI	SPGSLE	SPGSEN	SPGSGC	SPGSIN	SPGSLV	SPGSPM
S&P 500	1.000	0.885***	0.421***	0.211***	0.276***	0.234***	0.185*	-0.019***	0.227***	0.103***	0.000
MSCI	0.885***	1.000	0.622***	0.286***	0.395***	0.335***	0.240***	0.075***	0.349***	0.132***	0.102***
MSCIEM	0.421***	0.622***	1.000	0.263***	0.390***	0.322***	0.208***	0.122***	0.360***	0.102***	0.149***
GSCI	0.211***	0.286***	0.263***	1.000	0.914***	0.924***	0.966***	0.259***	0.389***	0.177***	0.276***
DJCI	0.276***	0.395***	0.390***	0.914***	1.000	0.968***	0.828***	0.396***	0.660***	0.209***	0.426***
SPGSLE	0.234***	0.335***	0.322***	0.924***	0.968***	1.000	0.803***	0.349***	0.547***	0.290***	0.373***
SPGSEN	0.185***	0.240***	0.208***	0.966***	0.828***	0.803***	1.000	0.187***	0.271***	0.097***	0.200***
SPGSGC	-0.019***	0.075*	0.122***	0.259***	0.396***	0.349***	0.187***	1.000	0.278***	0.031***	0.989***
SPGSIN	0.227***	0.349***	0.360***	0.389***	0.660***	0.547***	0.271***	0.278***	1.000	0.100***	0.308***
SPGSLV	0.103***	0.132***	0.102***	0.177***	0.209***	0.290***	0.097***	0.031***	0.100***	1.000	0.039***
SPGSPM	0.000	0.102***	0.149***	0.276***	0.426***	0.373***	0.200***	0.989***	0.308***	0.039***	1.000

Table 3. Simple correlation matrix based on the Pearson Correlation Coefficient with significance level for the daily returns between January 1, 1991 and September 15, 2023
(* = significance level of 0.01, ** = significance level of 0.05, * = significance level of 0.1)**

To already gain insight into correlations, a simple correlation matrix with the Pearson Correlation Coefficients was computed (see Table 3). The large majority of coefficients are significant at a level of 1%. Concerning stock indices, the S&P 500 and MSCI show a relatively high correlation, whereas the MSCIEM has a lower correlation with the first two. One may argue that this could influence the correlation with commodities and, therefore, whether commodities add value to the stock portfolio. For the S&P 500, the correlation with composite commodity indices is low but positive (ranging from 0.211 to 0.276).

Regarding individual commodity classes, the correlations tend to be even lower. This is in line with the findings of other scholars, who report low, positive correlations of equity and commodity returns for several subsamples and return periods (daily, weekly, monthly) (see Bessler & Wolff, 2015; Büyüksahin et al., 2009; Daskalaki & Skiadopoulos, 2011). For the SPGSGC, the coefficient is negative, though it is only significant at a level of 10%. Similarly, Bessler & Wolff (2015) state that the only insignificant correlation they find for the S&P 500 and a commodity subclass is with the precious metals index, leading them to conclude that precious metals are “best suited to complement a stock-only portfolio”

(Bessler & Wolff, 2015, p. 6). The MSCI as well as MSCIEM demonstrate higher correlations with commodities than the S&P 500. This might be attributed to United States focus of the S&P 500 – the MSCI and MSCIEM comprise stocks of economies, in which commodities have a higher importance.

As of cross-class commodity indices, it should be noted that they all correlate with each other with coefficients higher than 0.9. Additionally, not only the GSCI but also the DJCI and SPGSLE correlate strongly with the SPGSEN. The SPGSEN does, however, only demonstrate low correlation with the other commodity classes. Consequently, it may be assumed that also the DJCI and SPGSLE rely too greatly on energy assets and do not accurately depict the complete commodity market. While the high correlation of the GSCI and the SPGSEN is vastly discussed in literature (Bessler & Wolff, 2015; Büyüksahin et al., 2009; Daskalaki & Skiadopoulos, 2011; Lombardi & Ravazzolo, 2016), the DJCI and SPGSLE correlation with the SPGSEN has remained in the background.

Regarding the commodity subindices, the correlations among each other are low, but positive. This has also been discovered by other studies in the field (Ali et al., 2020; Daskalaki & Skiadopoulos, 2011; Erb & Harvey, 2006) and speaks for the heterogeneity of commodities, which makes it necessary to consider them individually as portfolio additions. The only exception is the high correlation of the SPGSGC with the SPGSPM with a coefficient of 0.989. This can be explained by the fact that gold is a precious metal, leading to common holdings of the SPGSGC and SPGSPM as well as a reciprocal impact of movements in the gold and precious metals market.

Moreover, the SPGSLV has the lowest correlation with other commodities with coefficients from 0.031 to 0.100. In contrast to these results, Büyüksahin et al. (2009) suggest a possible common economic force influencing the prices of most commodities due to an apparent strong, positive correlation of the S&P GSCI Non Energy index (SPGSNE) with the other subindices. Given that this thesis found no high correlations among the individual commodity class subindices, this indicates that they do not all share the same influential economic variable with the SPGSNE. The most probable reason for the deviation from Büyüksahin et al. (2009) are the different sample periods.

3.3 DCC GARCH Framework

To evaluate the correlation between commodities and stocks, the majority of research proposes the use of a multivariate GARCH model. In particular, the Dynamic Conditional Correlation (DCC) Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model by Engle (2002) has been used (see Büyüksahin et al., 2009; Chunhachinda et al., 2019; Creti et al., 2013; Junttila et al., 2018; Lombardi & Ravazzolo, 2016). The methodological goal of a DCC GARCH model can be described as modelling "the evolution of the relationship over time" (p. 10).

Though simple correlation measures, including rolling historical correlations and exponential smoothing techniques, are common in literature, they lack important attributes (Büyüksahin et al., 2009). For example, rolling historical correlations consider the time-varying nature of correlations by only including a certain time window with length k. Nonetheless, the approach is afflicted by assigning an equal weight to all observations in the specified window and zero weight to observations prior to the window. Therefore, window-length determination becomes a crucial issue since the window has to include all relevant and exclude all irrelevant information (Büyüksahin et al., 2009). Exponential smoothing approaches address this problem by assigning declining weights to older observations based on the parameter λ. This, however, creates the challenge of choosing the right smoothing parameter (Büyüksahin et al., 2009).

Moreover, Büyüksahin et al. (2009) point out that the most significant disadvantages of both methods are that they do not adequately consider changes in volatility as expressed in "the sensitivity of the estimated correlation to volatility changes restricts inferences about the true nature of the relationship between variables. Since the estimated correlations are subject to volatility shocks, interpreting these correlations becomes more difficult especially during high volatility periods." (Büyüksahin et al., 2009, p. 11). Similarly, Lombardi and Ravazzolo (2016) argue that the shortcoming of rolling windows is how they spread the impact of extreme market movements over time.

Due to these concerns, a more sophisticated model, such as the DCC GARCH, is necessary to adequately portray the correlation between stocks and commodities. This is the model employed in this thesis. In the following, the estimation of such a model is explained step-by-step.

The GARCH framework allows to model non-constant variance, which depends upon past variance values (Bollerslev, 1986). In a GARCH (p, q) model for conditional variance, forecasts depend on a (non-negatively) weighted sum of past squared residuals (from some conditional mean function model) and past variance forecasts:

$$\sigma_{t+1|t}^2 = \bar{\omega} + \sum_{i=1}^{p} \alpha_i \varepsilon_{t-i+1}^2 + \sum_{j=1}^{q} \beta_j \sigma_{t-j+1|t-j}^2 \tag{2}$$

where ω is a constant or unconditional variance, $\beta_j \sigma_{t-j+1|t-j}^2$ equal to the last period's forecast, and $\alpha_i \varepsilon_{t-i+1}^2$ the last period's news. It is necessary and sufficient for covariance stationarity that $\omega > 0$, $\alpha_1; \alpha_2; \ldots; \alpha_p \geq 0$, and $\beta_1; \beta_2; \ldots; \beta_q \geq 0$, and $\sum_{i=1}^{p} \alpha_i + \sum_{j=1}^{q} \beta_j < 1$ (Bollerslev, 1986).

The standard GARCH model can only be used for univariate time series. Hence, in the following, the multivariate DCC GARCH model will be implemented. The great advantage of the DCC GARCH model is that the time-varying interdependence of the volatility of one variable and the volatility of another variable can be studied. The DCC model follows a two-step approach (Engle, 2002). First, the time-varying variances of the univariate time series are estimated using a standard GARCH model. Secondly, the time-varying correlation matrix is calculated based on the standardized residuals from the univariate GARCH models (Engle, 2002).

Formally, a $n \times 1$ vector of the normally distributed return series r_t of n assets with a mean of zero and covariance matrix H_t is assumed:

$$r_t \sim N(0, H_t) \tag{3}$$

$$H_t = D_t, R_t, D_t \tag{4}$$

where H_t is the conditional covariance matrix, R_t is the time varying correlation matrix, and D_t is a diagonal matrix of time-varying conditional standard deviations given by $D_t = diag\sqrt{E_{t-1}(r_{i,t}^2)} = diag\sqrt{h_{i,t}}$. As shown in equation 3, using a standard GARCH model in the first step, the diagonal matrix of time-varying conditional standard deviations is given by the following equation:

$$h_t^2 = \alpha_0 + \sum_{i=1}^{p} \alpha_i \varepsilon_{t-i}^2 + \sum_{i=1}^{q} \beta_j h_{t-i}^2 \tag{5}$$

where h_t^2 represents the time-varying conditional variance of the according variable. After the first step of estimating the univariate volatility models for every variable, the standardized residuals $\varepsilon_{it} = \frac{r_{it}}{\sqrt{h_{it}}}$ can be used to estimate the dynamic evolution of the correlation, which is defined as:

$$Q_t = (1 - \alpha - \beta)\bar{P} + \alpha \varepsilon_{t-1} \varepsilon_{t-1}^{'} + \beta Q_{t-1} \tag{6}$$

where $\bar{P} = E\left[\varepsilon_{t-1} \varepsilon_{t-1}^{'}\right]$ is the unconditional correlation matrix of the standardized residuals. $\alpha + \beta < 1$ is necessary for mean reversion in the model and indicates that the conditional correlation is not constant over time. While α measures the short-term volatility impact, meaning the persistency of the standardized residuals from the previous period, β measures the long-term effect of a shock impact on the conditional correlations, meaning the persistency of the conditional correlation process.

Followingly, the time-varying correlation matrix P_t can be written as:

$$\bar{P} = diag\{Q_t\}^{-1} Q_t diag\{Q_t\}^{-1} \tag{7}$$

where $diag\{Q_t\}^{-1}$ is an inverted diagonal matrix containing the square root of diagonal elements of Q_t. Formally, this means that:

$$diag\{Q_t\}^{-1} = \begin{matrix} 1/\sqrt{q_{11t}} & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 1/\sqrt{q_{nnt}} \end{matrix} \tag{8}$$

Given the normality assumption, the likelihood function can now be used with the maximum likelihood estimation to estimate the parameters of the model:

$$L = -\frac{1}{2} \sum_{t=1}^{T} (nlog(2\pi) + log|H_t| + r_t^{'} H_t^{-1} r_t) \tag{9}$$

$$= -\frac{1}{2}\sum_{t=1}^{T}(nlog(2\pi) + 2log|D_t| + r_t^{'}D_t^{-1}D_t^{-1}r_t - \varepsilon_t^{'}\varepsilon_t + log|R_t| + \varepsilon_t^{'}R_t^{-1}\,\varepsilon_t)$$

Lastly, the time-varying conditional correlations can be determined with the following equation:

$$p_{i,j,t} = \frac{q_{i,j,t}}{\sqrt{q_{ii,t}q_{jj,t}}} \tag{10}$$

where $q_{i,j,t}$ is the covariance between the variables i and j at time t, and $q_{ii,t}$ as well as $q_{jj,t}$ are the conditional variance estimates of i and j at time t, meaning they are the diagonal elements in the covariance matrix Q_t (Engle, 2002).

4 Discussion of Results

This section reports the findings. I first estimate the DCC GARCH model, discussed in Section 3.3., using the full sample of daily return data from January 1, 1991 to September 15, 2023, and evaluate the model parameters. Subsequently, the development of the correlation of the S&P 500 and the GSCI is analyzed with a focus on the relationship during the GFC, the COVID Market Crash, and after the Russian invasion in Sections 4.2 and 4.3. On the basis of these results, it is investigated if commodities can be classified as hedges, diversifiers, or safe havens in Section 4.4. Lastly, in Section 4.5, the correlation of the S&P 500 and the commodity subindices is analyzed. The DCC GARCH model is additionally estimated for alternative commodity and stock indices for the sake of robustness in Section 4.6.

4.1 Model Estimation

	α	p-value	β	p-value
GSCI	0.01962	0.00034***	0.97726	0.00000***
DJCI	0.01808	0.00001***	0.97855	0.00000***
SPGSEN	0.02054	0.00566***	0.97593	0.00000***
SPGSGC	0.02482	0.00001***	0.96069	0.00000***
SPGSIN	0.01011	0.00005***	0.98763	0.00000***
SPGSLE	0.01543	0.00001***	0.98209	0.00000***
SPGSLV	0.00241	0.20619	0.99595	0.00000***
SPGSPM	0.02489	0.00002***	0.96186	0.00000***

Table 4. α and β parameters and their according p-values for the DCC GARCH models of commodity indices and the S&P 500 (* = significance level of 0.01, ** = significance level of 0.05, * = significance level of 0.1)**

The parameters of the standard GARCH models of the univariate time series are not significant in all cases[1]. This does, however, not inhibit the fitness of the model. As discussed in Section 3.3, the more important parameters are the Alpha and Beta of the DCC GARCH model, which are listed in Table 4 for the correlations between the commodity indices and the S&P 500. Alpha is measuring how much of the correlation estimation for the two time series depends on recent shocks, while Beta measures how

[1] A table displaying all parameters of the standard GARCH models and their p-value can be found in the appendix.

much of the correlation is persistent. The condition $\alpha + \beta \leq 1$, implying stationarity and mean reversion, is fulfilled in all cases.

Alpha and Beta are significant for all pairwise correlations, except for the Alpha of the SPGSLV. This implies that the current correlation of the S&P 500 and the SPGSLV is influenced by all past correlations to the same extent instead of an unproportional impact of the most recent correlation estimate. In other words, the correlation between the S&P 500 and the SPGSLV is very stable.

The fact that the sum of Alpha and Beta is close to one for all indices, shows that the volatility is highly persistent for all commodity (sub)indices. This is in line with results of Creti et al. (2013). Concluding, the table shows that the DCC GARCH model is appropriate for the use in this thesis due to the high significance of parameters, and that the correlation of the S&P 500 and commodities is highly persistent.

4.2 Correlation of the S&P 500 and GSCI during the Financialization and Global Financial Crisis

Unlike traditional models that assume constant correlations, the DCC GARCH model captures the time-varying interdependence of the volatility of one variable and the volatility of another variable. In this way, it accounts for the fact that correlations between assets can change over time. Thus, a DCC GARCH model allows to study the dynamics of the correlation. In the following, the correlation of the S&P 500 and the GSCI, as well as its development during the financialization in the 2000s, including the GFC, is analyzed.

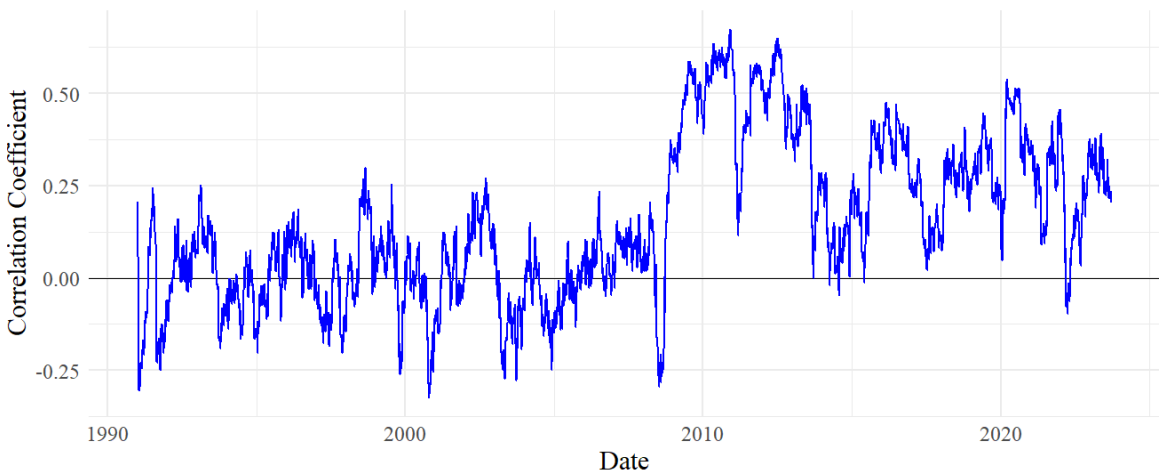

Figure 1. The time-varying correlation coefficient of the S&P 500 and GSCI between 1991 and 2023

As demonstrated in Figure 1, the dynamic correlation of the S&P 500 and the GSCI is substantially volatile. For the years from 1991 to 2008, the correlation fluctuated between approximately -0.25 and 0.25, and was characterized by moderately evenly distributed oscillations above and below zero. Some papers report a mean reversion around zero for this time period (Junttila et al., 2018; Lombardi & Ravazzolo, 2016; Nguyen et al., 2020). During this time, the coefficient remained almost always below the unconditional Pearson coefficient of 0.211, estimated in Section 3.2. The recession at the beginning of the 90s and the burst of the Dotcom Bubble do not appear to have had any impact on the correlation.

This trend did, however, change after a low at the beginning of July 2008 during the GFC. The correlation became positive at the end of September and has, since then, increased to all-time maximum values as well as it has remained positive for most of the time. Therefore, it can be stated that the correlation has experienced a shift towards higher, positive values. In fact, the highest coefficient value between 2008 and 2023 is more than twice the highest value between 1992 and 2008.

Concerning the financialization during the 2000s, the observance of the coefficient plot shows that the correlation began rising in 2005, though the trend was disrupted by the strong decrease in 2008. The more significant upward shift took place after 2008. There is a large body of academic research reaching the same findings regarding the development of the correlation beginning in 1991 and extending to the post-GFC time (Creti et al., 2013; Lombardi & Ravazzolo, 2016; Nguyen et al., 2020).

The start of financialization has frequently been dated to 2004 (Hamilton & Wu, 2014; Irwin & Sanders, 2011; Tang & Xiong, 2012). Irwin & Sanders (2011), for example, conduct statistical tests confirming that the correlation of commodity and stock returns increased significantly after 2004. Before 2004, investors seeking commodity investments followed the expensive practice of acquiring physical commodities or commodity related equities (Tang & Xiong, 2012). This changed in the mid-2000s, when index investments gained popularity. Given that the financialization was driven by index investments, it should be mentioned that off-index commodity investments have been found to be less affected (Tang & Xiong, 2012). For this reason, the results of this thesis should only with caution be generalized to off-index commodity instruments.

It is not possible to define one clear reason for the exact timing of the financialization (Carmona, 2015). The regulatory conditions for the financialization were set by the passing of the Commodities Futures Modernization Act in 2000. The act enabled commercial banks, as well as hedge funds, to invest more freely in commodity derivatives by excluding "the extra-exchange derivatives market from the supervision of the Commodities Futures Trading Commission (CFTC) and Securities and Exchange Commission (SEC)" (Włodarczyk & Szturo, 2018, p. 99). This regulatory innovation in combination with the publication of numerous studies (e.g. Erb & Harvey, 2006; Greer, 2000) highlighting the diversification benefits of commodities presumably heightened investors' interest (Tang & Xiong, 2012; Włodarczyk & Szturo, 2018). Tang & Xiong (2012) further suggest that the equity market collapse following the Dotcom Bubble led institutional investors to consider commodities as possible diversification instruments.

Nonetheless, it is notable that the impact of financialization can primarily be seen after the GFC (Ali et al., 2020; Basak & Pavlova, 2016; Büyüksahin et al., 2009). Before delving further into the interplay of the GFC and financialization, the development of the correlation during the GFC has to be explored. The GFC can be divided into two phases (Mishkin, 2011). The first one was the Subprime Mortgage Crisis from August 2007 to August 2008. Afterwards, in mid-September 2008, the crisis intensified and turned into a global crisis (Mishkin, 2011).

The Subprime Mortgage Crisis was more limited and originated from losses in subprime residential mortgages, constituting only a small sector within the US financial industry (Mishkin, 2011). During this time, the GDP still rose and institutions, such as the Congressional Budget Office, only forecasted a mild recession (Congressional Budget Office, 2008; Mishkin, 2011). The majority of public as well as private observers shared the consensus that the worst was over, the crisis could be contained, and the losses "seemed manageable" (Mishkin, 2011, p. 52). In September 2008, however, four major events revealed the severity of the situation and spread it to other regions.

First, the investment bank Lehman Brothers filed for the greatest bankruptcy in US history on September 15 (Mishkin, 2011). This was followed by the collapse of the AIG, a run on the market money Reserve Primary Fund, and the Congress's disapproval of the Troubled Asset Relief Plan, which would have given the US Treasury extensive access to resources for saving troubled financial institutions (Mishkin, 2011). These events "showed that risk

taking was far more extensive than markets had realized and the fragility of the financial system was far greater than most market participants could have imagined" (Mishkin, 2011, p. 55). Therefore, these events led to the transformation of the Subprime Mortgage Crisis into a grave, global financial crisis (Mishkin, 2011).

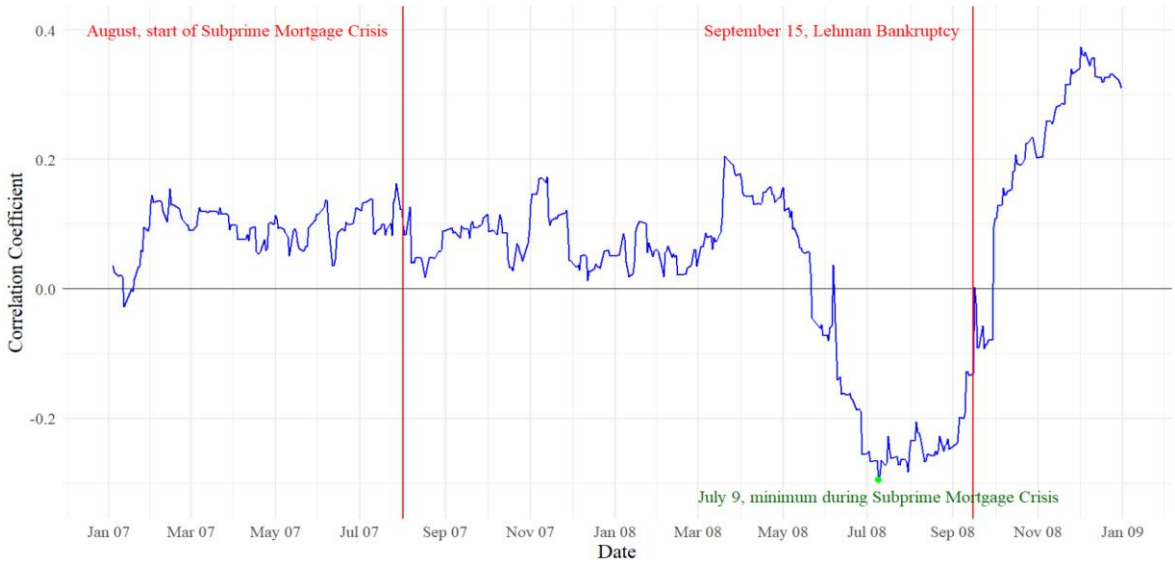

Figure 2. The time-varying correlation coefficient of the S&P 500 and GSCI during the GFC in 2007/8

Turning to the S&P 500 and GSCI correlation displayed in Figure 2, the indices were weakly positively correlated during 2007. Until April 2008, the correlation did not react to the Subprime Mortgage Crisis. Afterwards, the correlation coefficient began to decrease, and became negative in May. It reached the minimum of -0.296 in July, started to increase again, and turned positive at the end of September[2]. Thus, the correlation turned negative in the fourth quarter of the Subprime Mortgage Crisis and returned to be positive in the second, more severe phase of the GFC. Even more than that, the correlation increased to higher values than ever before. While the S&P 500 already declined in the Subprime Mortgage Crisis, commodity prices still followed a rising trend and only began to fall in the second phase of the GFC.

The increasing commodity prices in the first phase can be attributed to the commodity boom, beginning in 2002, which was driven by a change in supply-demand fundamentals (Nissanke, 2012). Newly emerging economies, such as China and India, heightened the demand of various commodities due to industrialization, increased food consumption, urbanization, and infrastructure building (Nissanke, 2012). Given that the Subprime

[2] Before turning positive on September 30, it jumps to zero for one day on September 16, the day following the Lehman Bankruptcy.

Mortgage Crisis was limited to the US, investors may have assumed that it would not influence the Asian countries and, therefore, not change the supply-demand pattern of the commodity markets. By September 2008, when the crisis showed its global repercussions, investors may have adjusted their demand forecasts for emerging markets. This would explain the drop in prices as well as the increase of correlation and suggest that commodity markets were not necessarily influenced by the stock market but by macroeconomic factors related to commodity demand.

Nissanke (2012) proposes that the fall of commodity prices can be attributed to the mentioned change in market sentiment to great extent but was also a result of the financialization in the sense of financial investors unwinding their positions independently from considerations of market fundamentals. According to Nissanke (2012), the crisis led financial investors to shift their investments from illiquid investments, such as commodity futures, to highly liquid assets, for example US treasury bonds, to reduce their portfolio risk during the crisis.

Furthermore, Creti et al. (2013) advance that the financialization could even have played a role in the rise of commodity prices during the Subprime Mortgage Crisis. This may be explained by the flight-to-quality phenomenon (Creti et al., 2013). When the risk of the stock market rises, investors seek diversification and frequently opt for refuge instruments. Gold is known as the most popular refuge instrument (Ali et al., 2020; Baur & Lucey, 2010; Junttila et al., 2018; Nguyen et al., 2020), though other commodity classes are affected as well. Creti et al. (2013) suggest that this could have been a price driver, next to the Asian demand.

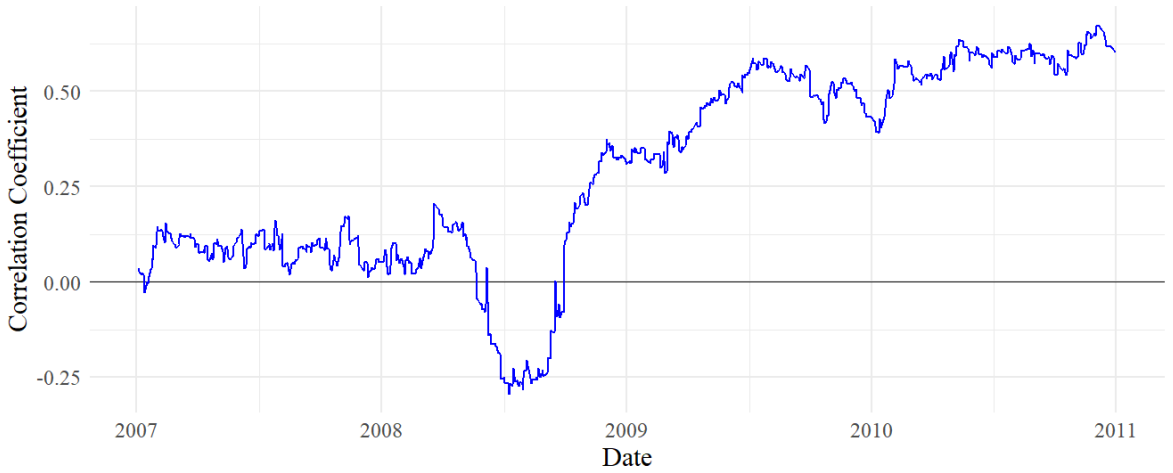

Figure 3. The time-varying correlation coefficient of the S&P 500 and GSCI during and shortly after the GFC

As already mentioned, and shown in Figure 3, the correlation shifted to a much higher level after 2008. This is in line with the majority of research on the financialization process, which suggests that it began in the mid-2000s, but intensified during and after the GFC (Basak & Pavlova, 2016; Büyükşahin & Robe, 2014; Tang & Xiong, 2012). Lombardi & Ravazzolo (2016) argue that it is not surprising that the correlation intensified in times of heightened volatility because it is probable that the prices were driven by the same influences, such as macroeconomic forecasts. This is confirmed by their formal test for a pattern breakpoint, based on Andrews & Ploberger (1994), finding the breakpoint on September 5 – shortly before the collapse of Lehman Brothers on September 15.

Nonetheless, the authors deem it remarkable "that seven years after the onset of the Great Recession, an instantaneous measure of correlation such as DCC still points to a substantially higher number compared to the pre-recession times" (Lombardi & Ravazzolo, 2016, p. 48). This demonstrates the likelihood of other forces, such as the financialization, next to the GFC, leading to the shift in correlations. In summary, most researchers suggest that the rise in correlation levels can be attributed to both, the crisis, and the phenomenon of financialization.

One possible connection would be that that the GFC has sustainably incentivized investors to hedge their equity portfolios with other asset classes and utilize their right to invest in commodity derivatives (Chunhachinda et al., 2019). On the other hand, commodities did not prove to be suitable hedges during the severe phase of the crisis. This speaks against the GFC as a motivation for hedging portfolios with commodities. The crisis may,

however, still have incentivized investors to simply diversify their portfolios with commodities.

Another explanation would be that the GFC showed financial investors that commodities do indeed behave similarly to equities in major economic crises. Since the GFC was the first market crisis after the passing of the Commodity Futures Modernization Act in 2000, it potentially was the first opportunity for investors to realize that commodities were not as disconnected from the equity market as they had assumed. If this was the case, this could have prompted them to execute their commodity trades similar to stock trades, resulting in a higher commodity-stock correlations.

Lastly, Junttila et al. (2018) suggest that the higher correlation could be a result of the zero-interest rate regime. They argue that low interest rates lead to low convenience yields[3], making the trade of futures contracts more attractive than the trade of the physical asset. This, in turn, could be visible in more investment activity in the commodity future markets since 2008. Speaking against this thesis is that the correlation has not returned to pre-2008 levels despite the rise of interest rates in 2022.

4.3 Correlation of the S&P 500 and GSCI during the COVID Market Crash and the Invasion of Ukraine

The GFC was followed by a decade of steady market growth, until the stability was challenged by COVID and the Russian invasion of Ukraine. After shortly elaborating on the correlation between 2010 and 2020, the development during these crises will be the focus of this section.

Looking back at Figure 1, it can be observed that, after the GFC, the correlation returned to a slightly lower level in 2014. Even though the level is not comparable to pre-financialization correlations[4], some scholars have called this phenomenon de-financialization, meaning a decrease of stock-commodity correlations due to a withdrawal of financial investors from the commodity market (Natoli, 2021). A de-financialization could have been driven by stricter regulation of the OTC commodity markets, leading investment banks, such as JP Morgan and Deutsche Bank, to downsize their commodity

[3] Convenience yield refers to the premium associated with holding a physical commodity instead of a derivative contract.
[4] As described in Section 4.2, the correlation in the 90s, for example, hovered around zero. The average dynamic, conditional correlation between 2014 and 2020 is 0.25. Between 2008 and 2014 it is 0.40.

trading operations in 2013 (Bianchi et al., 2018; Natoli, 2021). Natoli (2021), on the other hand, shows that although OTC markets have shrunk since 2008, financial investors are still active in trading exchange-traded derivatives with a peak of activity in 2020.

While Fry-McKibbin & McKinnon (2023) and Zhang et al. (2017) find no de-financialization effect, Adams et al. (2020) suggests a de-financialization of various commodity groups. Bianchi et al. (2018) advance that the energy market became more connected to the equity market since 2014, whereas other commodities de-financialized. In summary, the results from this stream of literature are still a question of debate and inconsistent (Natoli, 2021). Nonetheless, there is a consensus that the relationship has not reverted to its pre-financialization pattern, as is demonstrated in this thesis.

Figure 4. The time-varying correlation coefficient of the S&P 500 and GSCI during the COVID Market Crash and after the Russian invasion of Ukraine

Regarding the recent years of 2020 to 2023, an increase of correlation during the COVID Market Crash is visible. The decline of the global stock markets, which represented the most severe downturn since the GFC, began on February 20, 2020 (Song et al., 2022). Yet, information about the virus were already spread and first lockdowns established in January. This is reflected by the growth of the correlation at the end of January - the correlation began to increase shortly prior to the stock market crash, reaching its maximum with a correlation of 0.538 on March 17[5]. The end of the crash is frequently dated to April 7, with the recovery phase taking place from March 23 until the end of April (Capelle-Blancard & Desroziers, 2020). As demonstrated in Figure 4, it can be concluded that the correlation

[5] The S&P 500 reached its low on March 23.

remained at an exceptionally high level during the crash period and only normalized slowly in the months afterwards.

Notwithstanding, the correlation did not reach the level it had during the GFC. A decreasing trend from mid-2020 until mid-2021 can be observed, though the correlation remained positive. It, then, rose until 2022 and experienced a sharp decline at the beginning of 2022, turning negative in March. Afterwards, the correlation increased again, but stayed below 0.4.

The decline in 2022 may be attributed to the invasion of Ukraine by Russia on February 24, 2022 (Izzeldin et al., 2023). In fact, the decrease already began in January, and culminated in the first negative correlation lasting longer than 14 days since 2008. The correlation turned negative on March 10 and remained negative for 41 days. The minimum of -0.098 was reached on March 31. This covers the time during which the major immediate negative impact of the invasion on the equity market was visible. The S&P 500 fell by more than 7% in the days and weeks following the invasion but rebounded to its pre-war level within a month. Neely (2022) suggests that commodity prices returned to their more usual, more stable behavior by April. Figure 4 displays that the GSCI showed a negative correlation throughout the crisis and returned to positive values relatively quickly afterwards.

The downward trend in January could be attributed to the expectation of the geopolitical tensions, though it is also possible that other forces were driving the correlation during this time. Given that Russia and Ukraine are significant commodity export nations, it was foreseeable that the war would affect the correlation. For example, in 2021, Russia produced nearly 15% of the world's total supply of crude oil and Russian natural gas accounted for 40% of European gas demand (International Energy Agency, n.d.). Turning to Ukraine, the country was the seventh-largest wheat producer worldwide in 2022 (Khemakhem et al., 2023).

As an initial reaction the invasion, the GSCI price increased, while the S&P 500 declined. According to a report published by the World Bank (2022), the prices of wheat, European natural gas, and crude oil soared by more than 30% as a consequence of the war. Presumably, this was motivated by the expectable sinking supply of energy and agricultural commodities from Russia and Ukraine (Aizenman et al., 2023; Diaconaşu et al., 2023; Obi

et al., 2023). The war physically complicated the creation and extraction of commodities in both countries. A high number of farmers were not able to plant crops, and, especially at the beginning of the crisis, certain export routes, such as the Black Sea, were not accessible (Neely, 2022). Additionally, sanctions of both sides restricted the global commodity supply (Neely, 2022).

Subsequently, the GSCI price increased. Conversely, the stock prices declined due to investors' fear of a crash accompanied by cash necessity (Chowdhury & Khan, 2023). Moreover, the disruption of the commodity market was forecasted to have negative effects on companies' operations and cost structures (Neely, 2022). The expected reduction of profitability quickly began to be visible in the value of stocks (Basdekis et al., 2022). In particular, the soaring energy prices raised concerns, since they added to the already present inflation and had a negative impact on companies relying on energy consumption. This is why, for many investors, the invasion marked the possibility of recessionary times (Basdekis et al., 2022).

It is worth highlighting that the outcomes of COVID and the war were diametrically opposed – COVID resulted in an increase and a high, positive correlation, whereas the war resulted in a decrease and a negative correlation. This may be explained by the dissimilar nature and the role of commodities in the two crises. COVID was a completely unexpected, global crisis – it affected a variety of markets worldwide, especially through supply chain issues. Its effects on the stock market were enormous - for instance, the S&P 500's downturn activated the market-wide circuit breaker four times in March 2020, an event that had never occurred before in U.S. history (Naeem et al., 2022).

The Russian-Ukrainian war was more forecastable. Research suggests that the outbreak of the war was anticipated by investors a few days before it happened (Diaconaşu et al., 2023; Obi et al., 2023). Even earlier, geopolitical events, such as the Crimean annexation, indicated the possibility of tensions. Additionally, its impact was more localized and focused on certain industries, primarily those depending on energy and agriculture (Ben Amar et al., 2023). Particularly the role of commodities in the Russian and Ukrainian economies should be acknowledged and could potentially be a reason for the differing reactions to the two crises. In contrast to the price increase of the GSCI after the invasion, the index dropped in response to the COVID outbreak.

Due to the recency of the events, there is only a limited amount of academic literature on the impact of COVID as well as the Russian invasion on the correlation between stocks and commodities. There have not yet been papers published which investigate the correlation based on composite commodity indices. The existing research is focused on specific subclasses, namely oil, gas, and gold, as well as market connectedness networks rather than correlations.

Ben Amar et al. (2023) find an increase of connectedness during COVID and the Russian-Ukrainian War. They use a set of three commodity indices (energy, precious metals, and agricultural commodities). The effect is, however, stronger for COVID than for the war. Ben Amar et al. (2023) explain this with the more global nature of COVID events. Despite the growth of connectedness during the crises, Ben Amar et al. (2023) consider the market connectedness low enough for commodities to be viable "hedging instruments or diversifiers" (p. 16) for NATO stock portfolios. Their findings regarding the war are in line with other results. Several scholars constitute an increase of connectedness between the oil and stock markets during the war period (Adekoya et al., 2022; Khemakhem et al., 2023; Li et al., 2022; Okhrin et al., 2023). The same is found for gas and wheat by Khemakhem et al. (2023). Gold has been established as an effective safe haven by various authors (Adekoya et al., 2022; Khemakhem et al., 2023; Li et al., 2022).

Focusing on COVID, Liu et al. (2022) conclude an increase in correlations between the S&P 500 and various commodities (gold, oil, beverages, and wheat). Similarly, Hung & Vo (2021) and Naeem et al. (2022) draw on a wavelet approach and find that the returns transmission pattern between the S&P 500 and oil as well as gold thickened in the COVID-era. Farid et al. (2021) utilize the MCS-GARCH and Diebold & Yilmaz (2012) volatility index model and show that volatility connectedness among the S&P 500 and oil, gas, gold, as well as silver reached a peak during the COVID period. Despite the closer relationship of precious metals and the S&P 500 during the COVID crisis, all authors conclude that gold and silver could serve as valuable safe havens (Farid et al., 2021; Hung & Vo, 2021; Liu et al., 2022; Mujtaba et al., 2023; Naeem et al., 2022).

While the intensified relationship between stocks and commodities during COVID is resembled in the rise of correlation, the increment of connectedness during the Russian-Ukrainian war is inconsistent with findings of this thesis. Connectedness refers to the extent to which shocks or changes in one market can affect or be transmitted to another

market. High connectedness implies that shocks in one market are likely to have a significant impact on the other market, leading to a transmission of volatility or disturbances in either a positive or a negative way (Ben Amar et al., 2023).

Therefore, an increase of connectedness indicates that the correlation becomes stronger, meaning a higher absolute value of the correlation coefficient, regardless of the sign. As demonstrated in Figure 4, the correlation coefficient of the GSCI and the S&P 500 has decreased immensely in absolute value shortly after the invasion and has since then remained under its pre-war level. Thus, the connectedness between the GSCI and S&P 500 should have declined.

A potential source for this discrepancy is the use of different indices. As already mentioned, all published papers relate to specific commodities instead of a composite commodity index. Accordingly, it is possible that the connectedness of composite commodity indices differs from the connectedness of individual commodity indices. Considering that the cited authors used various approaches to estimate the connectedness (see e.g. Khemakhem et al., 2023 for TVP-VAR network connectedness; Ben Amar et al., 2023 for time and time-frequency connectedness based on the ForceAtlas2 algorithm; or Okhrin et al., 2023 for copula-based modeling), it is unlikely that the disparity is linked to a certain methodology. In the face of the robustness checks conducted within the scope of this thesis as well as the use of distinct stock indices in the mentioned works (see e.g. Khemakhem et al., 2023 for MSCI; Okhrin et al., 2023 or Adekoya et al., 2022 for S&P 500; and Ben Amar et al., 2023 for country-specific indices), it is also improbable that the divergence originates from the chosen stock indices. Ultimately, further research on the connectedness of the general commodity market and stock market as well as on the development of correlations during the war is necessary.

4.4 Classification as Diversifier, Hedge, and Safe Haven

The discussion will now turn to the question whether composite commodity indices can serve as diversifiers, hedges, or safe havens, based on the correlations estimated in Sections 4.2 and 4.3. To answer this question, it is necessary to have an overview of the impact of crises on the correlation. The findings from the previous subsections can be summarized as follows:

- GFC: At first, the weak, positive correlation did not react to the Subprime Mortgage Crisis. In the fourth quarter of the crisis, it turned negative. When the second, more severe and global phase of the GFC started, the correlation turned positive and increased to values higher than ever before.
- COVID Market Crash: The correlation increased, though not as strongly as during the GFC, and remained at the positive, above-average level during the complete time of the crash.
- Russian Invasion: The correlation decreased and turned negative for a longer time period for the first time since the GFC. Nonetheless, the negative correlation levels during the two mentioned crises are not comparable in value. The low in 2008 was -0.325, whereas it was -0.098 in 2022. The correlation in 2022 stayed negative until the stock markets recovered.

This leads to the conclusion that, based on the definitions of Baur & Lucey (2010), composite commodity indices can be considered as a diversifier, but not as a hedge or safe haven, for equity portfolios in the present. According to Baur & Lucey (2010), hedges should, on average, display a correlation equal to or below zero. Safe havens are characterized by correlations of zero or less during times of market distress, instead of on average. In contrast to this, diversifiers are assets which demonstrate a low positive correlation on average.

At least since the GFC, commodities are on average positively correlated with equities. It may be argued that the positive and negative correlations cancelled each other out before 2008, resulting in an average correlation of zero. Hence, commodities could have been utilized as a hedge prior to 2008. This would confirm the thesis of several authors who claim that the potential of commodities as a hedge has declined since the 2000s (Basak & Pavlova, 2016; Büyükşahin & Robe, 2014; Singleton, 2014; Tang & Xiong, 2012).

Regarding the correlation during times of market turmoil, the results are contradictory. While the relationship during the GFC was characterized by both negative and positive correlations, COVID lead to an increase in positive correlation and the Russian invasion caused a negative correlation. This may be connected to the gravity of the crises. The first phase of the GFC as well as the Russian invasion of Ukraine had a limited impact on the market, especially in comparison to the second phase of the GFC and COVID. The GSCI-S&P 500 correlation was negative during the firstly mentioned and positive in the latter

mentioned periods. Thus, it may be concluded that the correlation was negative during minor market downturns, and positive during major events. Subsequently, the condition for safe havens is only satisfied for small market crashes.

Nonetheless, commodities could be evaluated as diversifiers based on their correlations with equities since 2008. Given the average correlation of 0.329 of the S&P 500 and GSCI since October 2008, the condition of a positive (but not perfect) correlation may be considered fulfilled. It should still be taken into account that the maximum correlation is relatively high with 0.674. Consequently, they may be regarded as a risky diversification.

4.5 Correlation and Classification of Commodity Subclasses

As mentioned previously, commodities are a heterogenous asset class. This is why it is necessary to analyze the correlation of stocks with individual commodity subclasses. The following section revolves around the correlation of the S&P 500 with five different commodity groups[6]. Moreover, it will be examined if the individual subclasses can act as diversifiers, hedges, or safe havens, based on Baur & Lucey (2010).

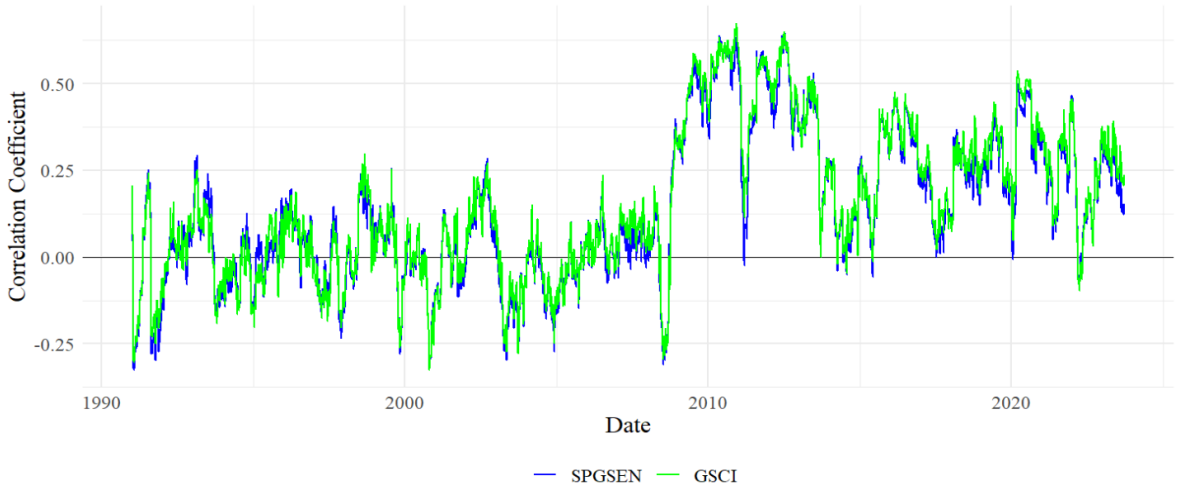

Figure 5. The time-varying correlation coefficient of the S&P 500 and SPGSEN (blue) in comparison to the coefficient of the S&P 500 and GSCI (green)

The correlation of the S&P 500 and SPGSEN moves relatively in line with the GSCI correlation. This may be attributed to the overweight of energy assets in the GSCI. Therefore, all findings regarding the GSCI are also valid for the SPGSEN. The same has

[6] Energy is represented by the SPGSEN index, livestock by the SPGSLV, industrial metals by the SPGSIN, precious metals by the SPGSPM, and gold by the SPGSGC.

been observed in studies with other time frames and methodologies (Büyüksahin et al., 2009; Junttila et al., 2018; Nguyen et al., 2020).

It should, however, be pointed out that individual energy subclasses may serve as better or worse hedges than energy indices. This is indicated by the research of Creti et al. (2013), who find that gas and electricity demonstrate significantly lower correlations with equities than oil. Especially electricity frequently displays a negative correlation with equities, suggesting that it may be a valuable hedge. In contrast to this, the SPGSEN would be classified as a diversifier based on the results of this thesis.

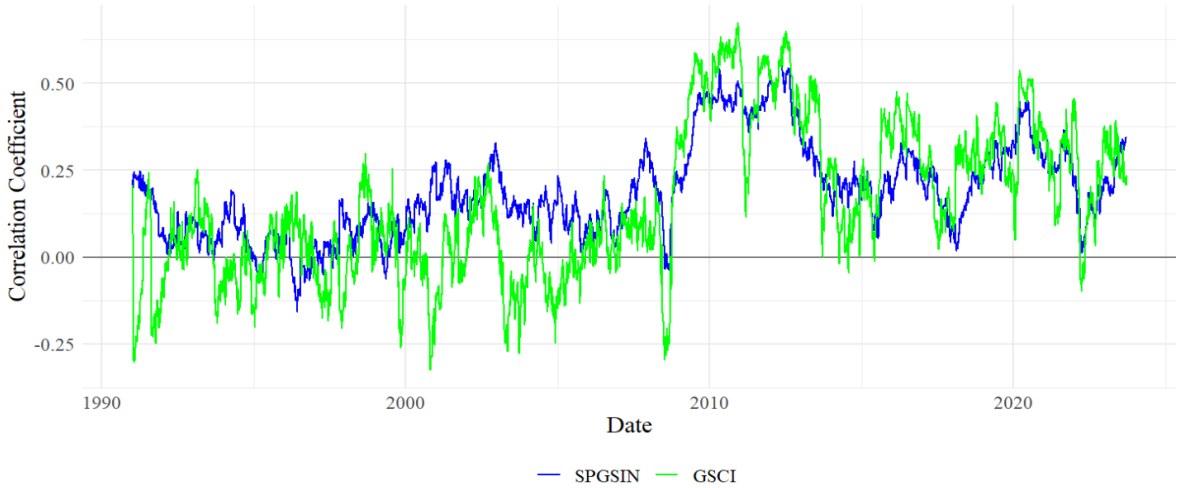

Figure 6. The time-varying correlation coefficient of the S&P 500 and SPGSIN (blue) in comparison to the coefficient of the S&P 500 and GSCI (green)

As depicted in Figure 6, the SPGSIN more frequently exhibits positive correlations during the 90s and 2000s, when compared to the GSCI. There are several points in time for which the correlation of the SPGSIN is positive but the correlation of the GSCI negative. As a consequence, the SPGSIN can, in contrast to the GSCI, not be considered as a hedge prior to 2008. The average correlation of the SPGSIN with the S&P 500 over this time period equals 0.108, wherefore the condition of a correlation equal to or below zero is not fulfilled. This confirms the results of Garcia-Feijoo et al. (2012). Since 2008, the GSCI correlation levels have often been higher than the SPGSIN correlation. On the other hand, downward swings of the GSCI are stronger.

As of crises, the SPGSIN tends to have fewer negative correlations. This constitutes the SPGSIN as an even more inadequate safe haven than the GSCI. The GSCI could have served as a safe haven in minor crises, whereas the SPGSIN had positive correlations in minor as well as major crises. Similar to the GSCI, the SPGSIN could be viewed as a

diversifier, with an average correlation of 0.188 throughout the complete sample period. The lower average as well as maximum correlation since 2008 positions the SPGSIN as a better diversifier than the GSCI.

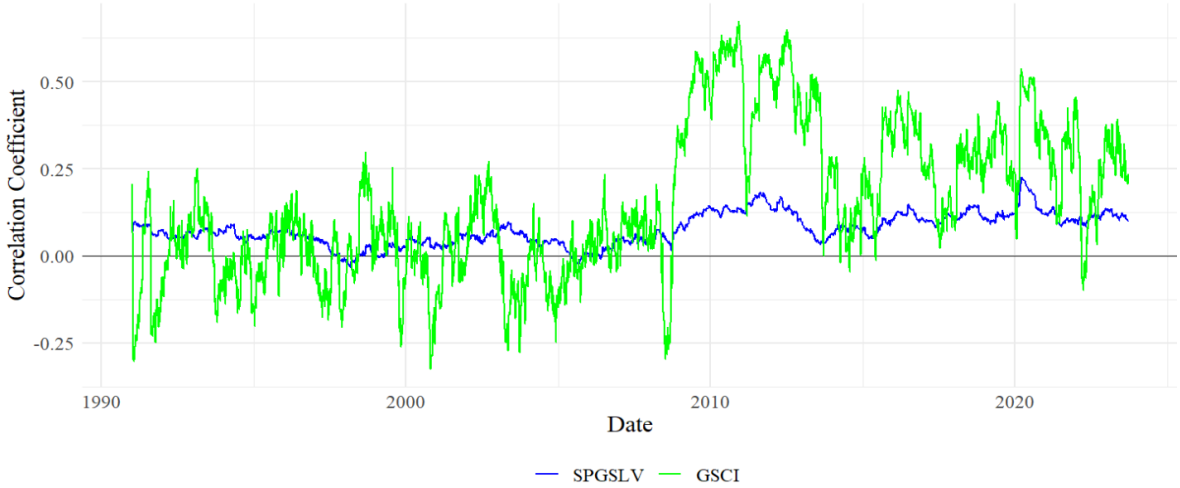

Figure 7. The time-varying correlation coefficient of the S&P 500 and SPGSLV (blue) in comparison to the coefficient of the S&P 500 and GSCI (green)

Looking at the SPGSLV in Figure 6, the correlation differs greatly from the GSCI correlation. The SPGSLV almost constantly maintains a low, positive correlation with the S&P 500 with a small, sustainable shift into the positive direction during the 2000s. This leads to an average correlation of 0.0753 and a maximum of 0.227. The correlation did neither show a strong reaction to the GFC nor to the Russian invasion. It increased slightly during COVID, though it returned to its normal level very quickly.

Given the low average as well as maximum correlation, it may be viewed as the best diversifier out of all analyzed commodity indices. It may even be suggested that the SPGSLV could serve as a hedge or safe haven, depending on the time horizon and on how broadly the term "uncorrelated" is defined. This result confirms corresponding literature. For example, Silvennoinen & Thorp (2013) report that "meat and livestock futures are not very closely correlated with stock returns" (p. 54). Regarding research on portfolio simulation, it has been illustrated that, out of all commodity classes, the highest amount of variance can be reduced by livestock (Olson et al., 2017).

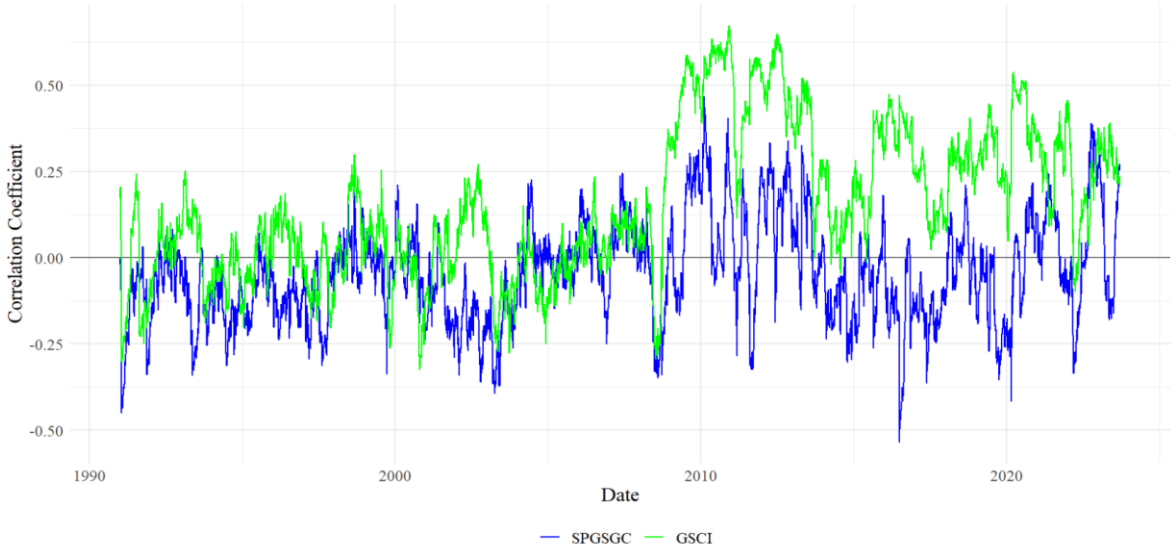

Figure 8. The time-varying correlation coefficient of the S&P 500 and SPGSGC (blue) in comparison to the coefficient of the S&P 500 and GSCI (green)

The SPGSGC is the only commodity index which has a negative average correlation with the S&P 500 - the average correlation equals -0.047. As depicted in Figure 8, the correlation was primarily negative until the mid-2000s. Then, it shifted to a higher level until it dropped, similar to the GSCI correlation, at the beginning of the GFC. In contrast to the GSCI correlation, the decrease was not followed by a sustainable increase. Instead, the correlation has been mostly negative since 2013. In summary, the SPGSGC could be evaluated as a hedge given the low negative average correlation. Thereby, the results support the findings of other studies (Ali et al., 2020; Junttila et al., 2018; Nguyen et al., 2020).

During the GFC, the SPGSGC served as a safe haven. Even though the correlation was mostly weak but positive during the Subprime Mortgage Crisis, it was primarily negative during the second phase of the GFC, when investors needed assets with safe haven properties the most. The same holds true for the COVID Market Crash. The correlation was negative when the crash began and stayed negative until the market started to recover. Though the correlation was slightly positive during the rebound phase and after (from April until June), it stayed below 0.1. As of the reaction to the Russian invasion, the correlation was negative at the time of the invasion and remained negative for the five months following the invasion. Therefore, gold also served as an excellent safe haven in this event.

This thesis corroborates numerous studies attesting the safe haven potential of gold. In particular, the works of Baur & Lucey (2010), Junttila et al. (2018), and Nguyen et al. (2020) can be consulted for further analyses on gold as a safe haven during the GFC. For research on the safe haven potential during the Russian-Ukrainian war, please see Adekoya et al. (2022), Ben Amar et al. (2023), Khemakhem et al. (2023), and Li et al. (2022). The COVID Market Crash is investigated by Kinateder et al. (2021), Lahiani et al. (2021), and Mujtaba et al. (2023).

Since gold is a precious metal, the SPGSGC moves relatively in line with the SPGSPM, wherefore all findings apply to the SPGSPM as well.

4.6 Robustness Checks Using Different Indices

To ensure the reliability of the results, several robustness checks were performed. They comprised all possible equity-commodity combinations with the S&P 500, MSCI, and MSCIEM as equity benchmarks and the GSCI, DJCI, SPGSLE, as well as the various subclasses as commodity benchmarks. The outcomes of these tests will be reported in the following subsection.

The DCC GARCH model built for the S&P 500 and DJCI shows the same patterns and correlation levels as the GSCI. The same applies to the SPGSLE, though the SPGSLE has marginally higher correlations. Thus, the use of alternative composite commodity indices confirms the robustness of the results. Evidently, the overweight of energy assets in the GSCI does not distort the findings. The results also remain valid when using the MSCI as the equity index. Small deviations include that the MSCI tends to have more extreme correlation values with the GSCI, as well as a more frequently positive correlations with the DJCI and SPGSLE.

Concerning the MSCIEM and GSCI, the correlations during the 90s and early 2000s tended to be closer to zero. It should be pointed out they have constantly been positive since 2003. During the GFC, the correlation dropped slightly at the beginning, but immediately began to rise after the Lehman collapse on September 15. The strong negative correlation, which can be seen with the S&P 500 in August 2008, is not visible with the MSCIEM. Though the Russian invasion led to a decrease of correlation between MSCIEM and GSCI, it neither resulted in a negative value.

All of these findings hold true for the MSCIEM and SPGSLE. For the DJCI, the correlation is even more frequently positive. For the complete sample period, the correlation with the MSCIEM was only negative for 46 days in 2000. Before 2007, the correlation with the MSCIEM was significantly higher than with the S&P 500. After the spike in the S&P 500 correlation due to the GFC, the two correlations stabilized at a comparable level. Nonetheless, the correlation with the MSCIEM remains marginally higher.

A possible explanation for the higher, positive correlations is that the growth of emerging economies is accompanied by a heightened demand for a broad range of commodities, such as energy or metals (Hamilton, 2009; Kilian, 2009; Tang & Xiong, 2012). For example, the seven largest emerging markets were responsible for "92 percent of the increase in metals consumption, 67 percent of the increase in energy consumption, and 39 percent of the increase in global food consumption" (Baffes et al., 2018, p. 2) between 2000 and 2018. This is visible in the closer relationship of commodities and emerging market stocks.

Ultimately, the robustness check with the MSCIEM implies that the value of commodities for stock portfolios differs for emerging and developed markets since the 2000s. For the S&P 500 and MSCI it was found that composite commodity indices acted as a safe haven during minor crises (see Section 4.4). In contrast to this, the positive correlation of the MSCIEM and GSCI since 2003 proves that they did not act as a safe haven for the MSCIEM. Apart from safe haven properties, the findings are the same for the MSCIEM and developed market stock indices. For both, emerging and developed market portfolios, commodities could have been used as a hedge during the 90s and could be used as a diversifier since then.

Robustness checks for the commodity subclasses based on the MSCI reveal relationships similar to the S&P 500 with all subclasses, the only difference being that all correlations are shifted to a slightly higher level. For the MSCIEM, deviations include that the correlation with the SPGSIN is almost constantly higher with fewer negative correlations and a smaller dip during the GFC. Nonetheless, this does not inhibit the diversifier potential of the SPGSIN for the MSCIEM. The correlation between the MSCIEM and SPGSGC displays more major discrepancies to the S&P 500. It has significantly more frequently positive values, leading to an average of 0.105. Accordingly, in contrast to the

results for the S&P 500 and MSCI, the SPGSGC does not serve as a hedge or safe haven for the MSCIEM.

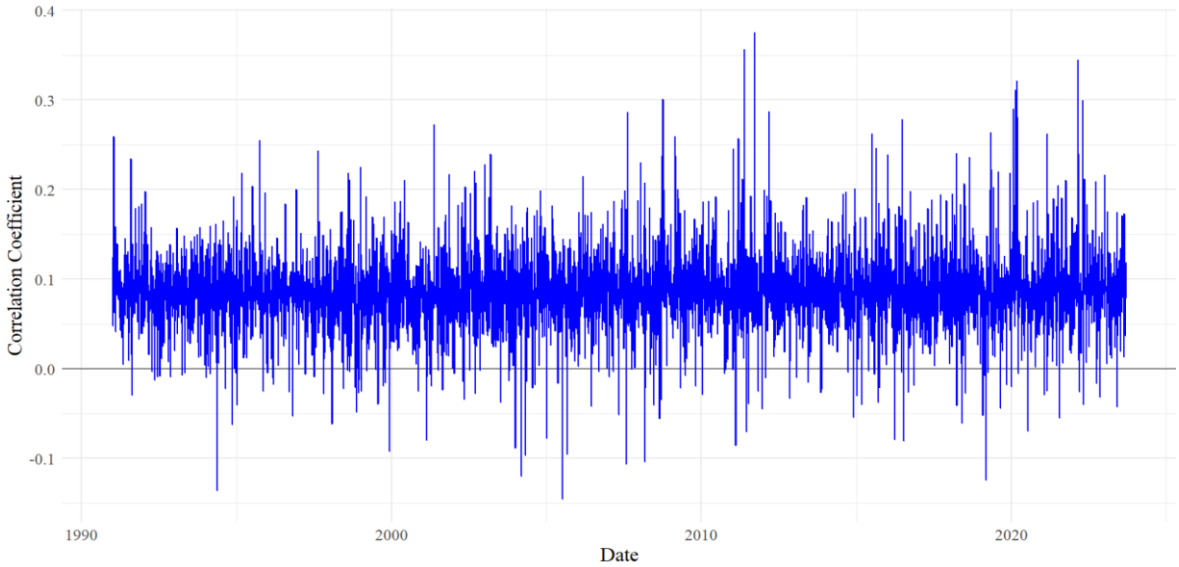

Figure 9. The time-varying correlation coefficient of the MSCIEM and SPGSLV

As demonstrated in Figure 9, the most unusual correlation is the MSCIEM's correlation with the SPGSLV. Throughout the complete sample period, the correlation shows a mean-reverting pattern around the mean of 0.0862. While the average does not significantly differ from the average correlation of the S&P 500 and the SPGSLV (0.0753), the pattern of the MSCIEM correlation is striking. The mean reversion is proven statistically by means of an Augmented Dickey Fuller-Test (see Dickey & Fuller, 1979), which is the most widely used test for mean reversion (Özel & Ilalan, 2018). This does, however, not change the value of the SPGSLV as a portfolio addition. Equal to the results for the S&P 500 presented in Section 4.5, the weak correlation positions the SPGSLV as an excellent diversifier and a potential hedge or safe haven, depending on how strictly the term "uncorrelated" is defined, for the MSCIEM.

5 Conclusion

The purpose of this thesis was to evaluate the potential of commodities as diversifiers, hedges, and safe havens for stock portfolios. To achieve this, the GSCI as well as its subindices were analyzed by means of their conditional, dynamic correlations with the S&P 500. Based on the correlations of the DCC-GARCH model, the commodity indices were classified according to Baur & Lucey (2010). As one of the first works examining the correlation of commodities and equities during the COVID Market Crash and the Russian invasion, this thesis adds essential value to the body of existing research.

The results show that composite commodity indices are not sufficiently negatively or un-correlated with equities to serve as hedges or safe havens for strong market downturns - they only are suitable diversifiers. Even their potential as diversifiers could be questioned given the average correlation of 0.329. This depends on how broadly the condition of a positive, but not perfect, correlation is defined (Baur & Lucey, 2010). Regarding safe haven properties, it should be pointed out that commodities and stocks were negatively correlated, thereby fulfilling the conditions for a safe haven, during minor crises.

In Section 4.2, it has been proven that the GFC, presumably in connection with financialization, had a major impact on the correlation. After the GFC, the correlation levels experienced a significant shift into the positive direction, marking closer links between commodity and equity markets. For example, commodities could be evaluated as a valuable hedge with an average correlation of approximately zero prior to 2008. Since then, they could only be used as diversifiers. The correlation was negative during the Subprime Mortgage Crisis, the first phase of the GFC, but turned positive when the crisis developed a more severe and global nature. This is an example of the influence of crisis severity on the adequacy of commodities as safe havens.

Section 4.3 showed that the economic downturns due to COVID and the Russian invasion led to different reactions. While COVID resulted in an increase of the already positive correlation, the Russian invasion was accompanied by a decline and slightly negative correlation values. This may be rooted in the special role of commodity exports for Russia or be seen as another example for the safe haven potential depending on crisis intensity.

Turning to commodity subindices in Section 4.5, the SPGSEN (energy), SPGSIN (industrial metals), and SPGSLV (livestock) could serve as diversifiers. Notably, the

SPGSLV has the lowest average correlation with 0.0753, constituting it as the best diversifier. Solely based on average correlation, the SPGSLV could be deemed the superior diversifier when compared to the GSCI. The SPGSGC (gold) and SPGSPM (precious metals) have low negative average correlations, making them effective hedges. During crises, both indices acted as safe havens.

For the sake of robustness, the DCC-GARCH model was computed for alternative commodity as well as equity indices in Section 4.6. These comprised the DJCI and SPGSLE as commodity indices and the MSCI and MSCIEM as equity indices. Neither the substitution of the GSCI with the DJCI and SPGSLE nor the substitution of the S&P 500 with the MSCI yielded different results.

Conversely, the MSCIEM has significantly more frequently positive correlations with the composite commodity indices as well as gold. This inhibits the safe haven, though not the diversification, potential of composite commodity indices for the MSCIEM. Similarly, gold does not serve as a hedge or safe haven for the MSCIEM. The deviations may be explained by the strong connection between the growth of emerging markets and commodity demand.

In summary, the value commodities add to stock portfolios depends on the specific commodities, equities and time period. Generally, composite commodity indices are inadequate hedges and safe havens, though they could serve as diversifiers. This is due to the higher correlation with equities since the GFC and financialization. Before the GFC, commodities were effective hedges. The use as a safe haven is possible during minor but not major market downturns.

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Appendix A GARCH Parameters and Index Price Development

A.1 Parameters of the Standard GARCH Models

	Estimate	Std. Error	t value	Pr(>|t|)
μ S&P 500	0.00062	0.00008	7.91484	0.00000***
Ω S&P 500	0.00000	0.00000	0.39371	0.69379
α S&P 500	0.11065	0.05141	2.15248	0.03136**
β S&P 500	0.87495	0.06024	14.52537	0.00000***
μ MSCI	0.00050	0.00018	2.86107	0.00422***
Ω MSCI	0.00000	0.00001	0.11775	0.90626
α MSCI	0.10909	0.19208	0.56798	0.57005
β MSCI	0.87849	0.19579	4.48687	0.00001***
μ MSCIEM	0.00061	0.00013	4.71648	0.00000***
Ω MSCIEM	0.00000	0.00001	0.34331	0.73136
α MSCIEM	0.10466	0.06382	1.63991	0.10102
β MSCIEM	0.88265	0.07046	12.52720	0.00000***
μ GSCI	0.00018	0.00014	1.26589	0.20555
Ω GSCI	0.00000	0.00001	0.10935	0.91292
α GSCI	0.07607	0.12970	0.58652	0.55752
β GSCI	0.92093	0.12788	7.20141	0.00000***
μ DJCI	0.00031	0.00012	2.68544	0.00724***
Ω DJCI	0.00000	0.00000	0.45175	0.65145
α DJCI	0.05655	0.03614	1.56484	0.11762
β DJCI	0.93615	0.03649	25.65377	0.00000***
μ SPGSEN	0.00041	0.00019	2.15128	0.03145**
Ω SPGSEN	0.00000	0.00000	1.23557	0.21662
α SPGSEN	0.09195	0.01373	6.69699	0.00000***
β SPGSEN	0.89904	0.01686	53.30953	0.00000***
μ SPGSGC	0.00000	0.00009	-0.00258	0.99795
Ω SPGSGC	0.00000	0.00000	0.09560	0.92384

α SPGSGC	0.04299	0.00816	5.26970	0.00000***
β SPGSGC	0.95601	0.00789	121.18160	0.00000***
μ SPGSIN	0.00008	0.00012	0.63474	0.52560
Ω SPGSIN	0.00000	0.00000	1.17121	0.24151
α SPGSIN	0.04768	0.01950	2.44497	0.01449**
β SPGSIN	0.94551	0.01987	47.58874	0.00000***
μ SPGSLE	0.00012	0.00009	1.40609	0.15970
Ω SPGSLE	0.00000	0.00000	0.43204	0.66572
α SPGSLE	0.05934	0.02738	2.16711	0.03023**
β SPGSLE	0.93764	0.02627	35.68836	0.00000***
μ SPGSLV	0.00011	0.00010	1.02821	0.30385
Ω SPGSLV	0.00000	0.00000	0.92782	0.35350
α SPGSLV	0.04717	0.02379	1.98272	0.04740**
β SPGSLV	0.93894	0.02616	35.89599	0.00000***
μ SPGSPM	0.00000	0.00009	0.00101	0.99920
Ω SPGSPM	0.00000	0.00000	0.14712	0.88304
α SPGSPM	0.04144	0.00560	7.39926	0.00000***
β SPGSPM	0.95700	0.00538	177.90193	0.00000***

Table 5. Parameters of the standard GARCH models and their according p-values for the commodity and stock indices (* = significance level of 0.01, ** = significance level of 0.05, * = significance level of 0.1)**

A.2 Price Development of Stock and Commodity Indices

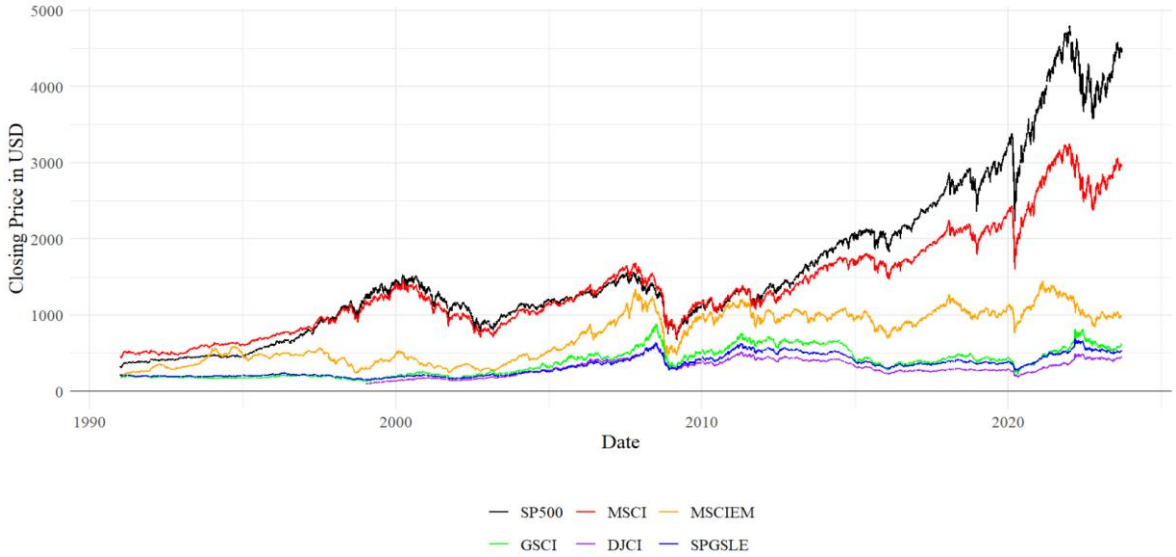

Figure 10. Price Development of Stock and Composite Commodity Indices between January 1991 and September 2023

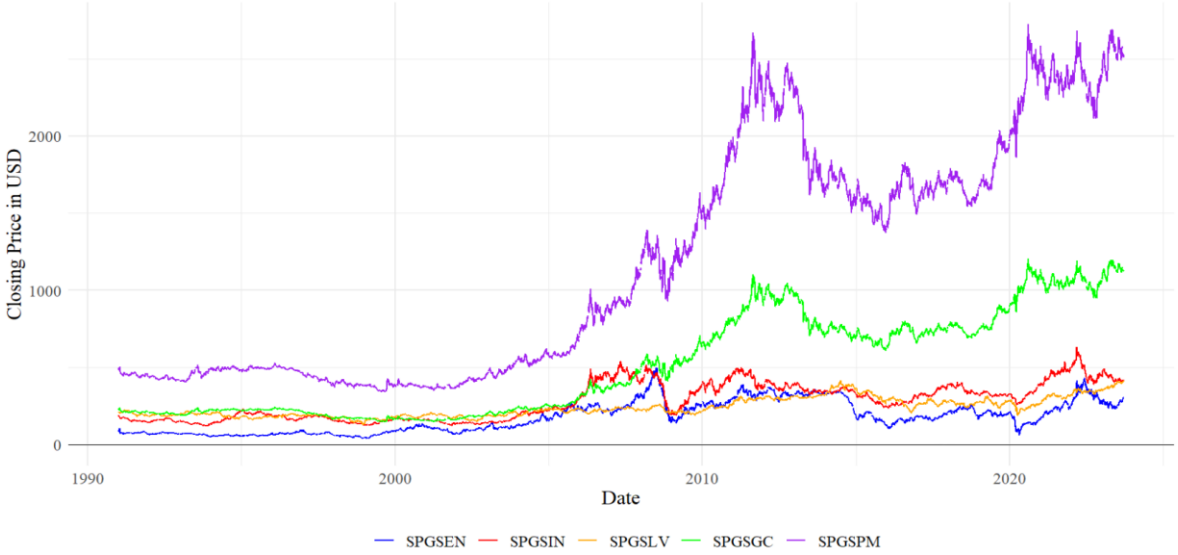

Figure 11. Price Development of Sub-Commodity Indices between January 1991 and September 2023

Declaration of authorship of an academic paper

I hereby declare that I have written this paper myself and used no other sources or resources than those indicated, have clearly marked verbatim quotations as such, and clearly indicated the source of all paraphrased references, and have observed the General Study and Examination Regulations of Reutlingen University for bachelor and master programmes, the specific regulations for study and examinations of my study programme, and the Regulations for Ensuring Good Academic Practice of Reutlingen University.

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Elon, 15.04.2024

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